Backpropagation e como ele ajusta os pesos em redes profundas para minimizar o erro.

Davi Duarte de Paula

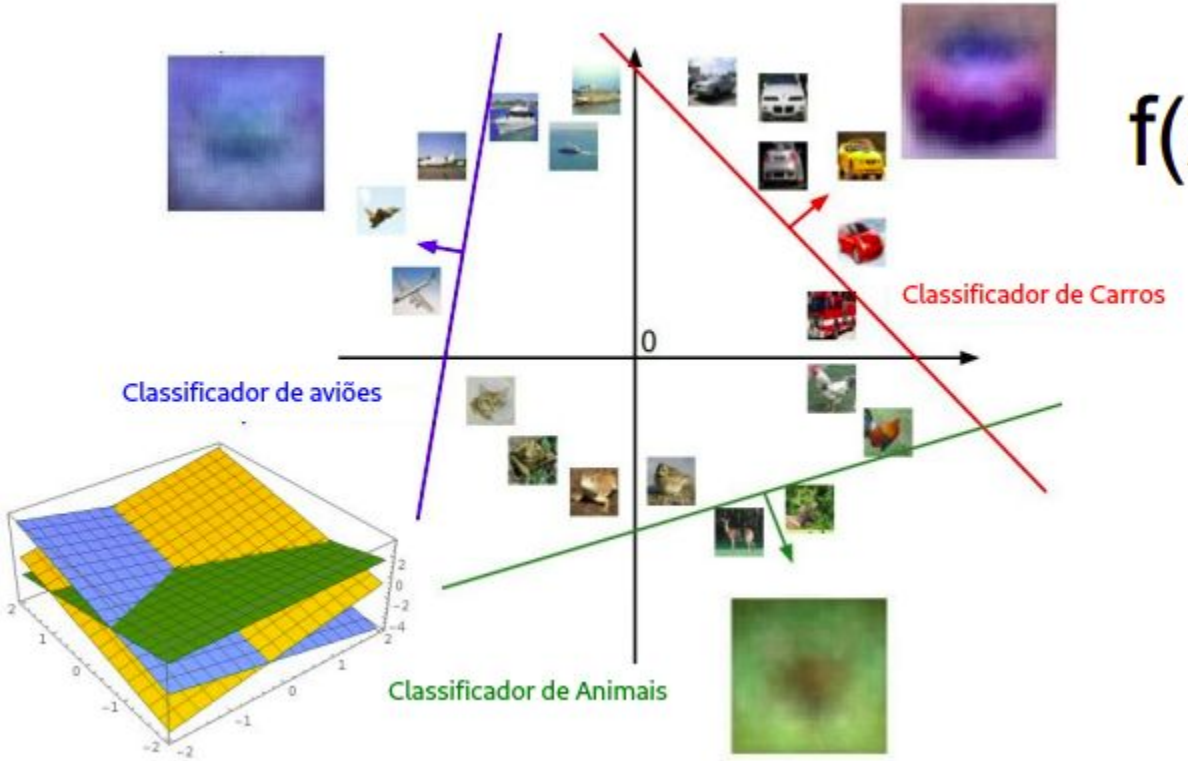
Fatec Rio Claro

Centro Paula Souza

Sumário

- Classificador Linear (Aulas passadas)
- Gradiente
- Gradiente Descendente
- Backpropagation
- Exercício Resolvido

Aulas passadas - Classificador Linear

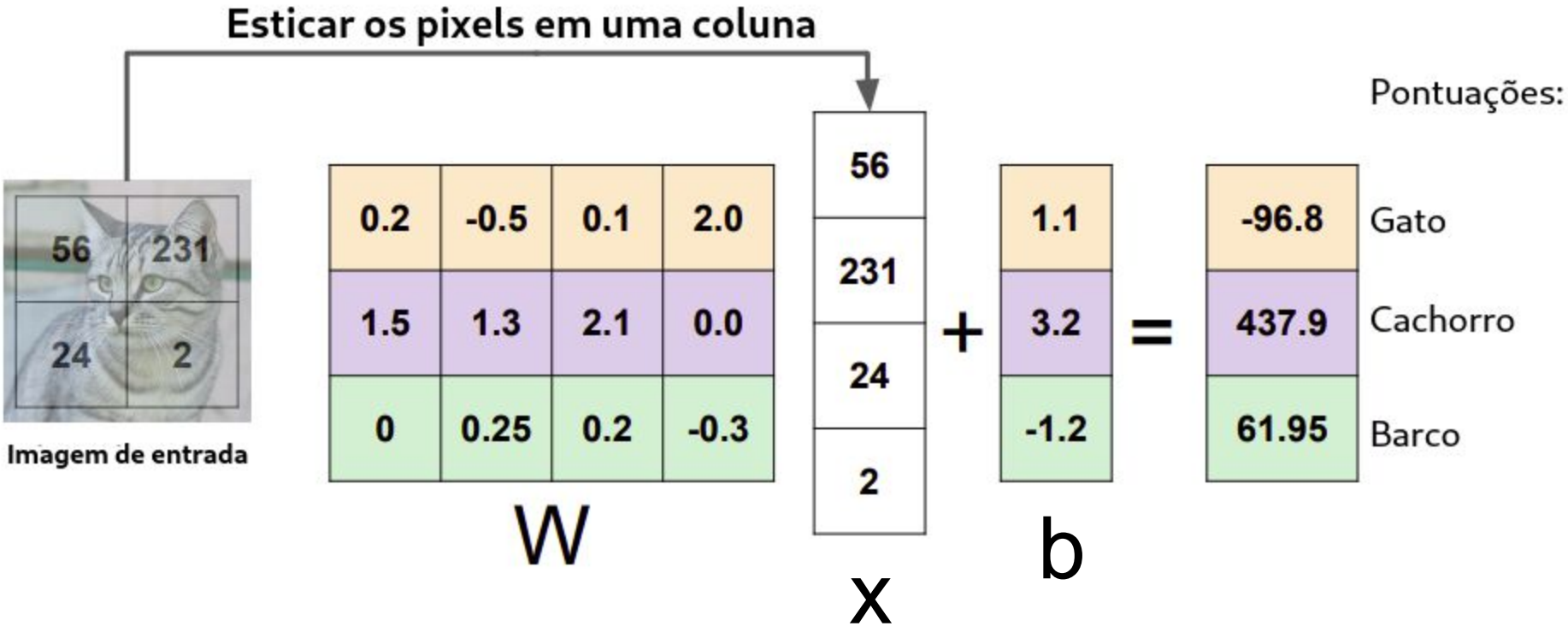


$$f(x,W) = Wx + b$$



Array de 32x32x3 números
(3072 números no total)

Aulas passadas - Classificador Linear



Aulas Passadas

- Classificador Linear

$$s = f(x; W) = Wx$$

- Função de custo Support Vector Machine (SVM)

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Aulas Passadas - Custo SVM

Suponha 3 amostras de treinamento e 3 classes.
Com algum valor de W os scores f(x, W) = Wx são:





Gato	**3.2**	1.3	2.2
Carro	5.1	**4.9**	2.5
Barco	-1.7	2.0	**-3.1**

Aulas Passadas - Custo SVM

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Abreviando $s = f(x_i, W)$

A função SVM L_i (para múltiplas classes)

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Abreviando $s = f(x_i, W)$

A função SVM (para múltiplas classes), tem a forma:

Q1 - O quê acontece se W for ruim?

Aulas Passadas - Custo SVM

Suponha 3 amostras de treinamento e 3 classes.
Com algum valor de W os scores f(x, W) = Wx são:





Gato	**3.2**	1.3	2.2
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Custos	2.9	0	12.9

Abreviando $s = f(x_i, W)$

A função SVM (para múltiplas classes), tem a forma:

Q2 - Como encontrar o melhor valor de W?

Otimização

Otimização: Gradiente



Otimização: Gradiente



Otimização: Gradiente

Em 1 dimensão, a derivada de uma função

$$\frac{df(x)}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Em múltiplas dimensões, o **gradiente** é o vetor das derivadas parciais ao longo de cada dimensão

Otimização: Gradiente usando Diferenças Finitas

W Atual

```
[0.34,
-1.11,
0.78,
0.12,
0.55,
2.81,
-3.1,
-1.5,
0.33,...]
```

Gradiente dW

```
[?,
?,
?,
?,
?,
?,
?,
?,
?,...]
```

Otimização: Gradiente usando Diferenças Finitas

W Atual	W + h (1ª dim)	Gradiente dW
[0.34,	[0.34 + **0.0001**,	[?,
-1.11,	-1.11,	?,
0.78,	0.78,	?,
0.12,	0.12,	?,
0.55,	0.55,	?,
2.81,	2.81,	?,
-3.1,	-3.1,	?,
-1.5,	-1.5,	?,
0.33,…]	0.33,…]	?,…]

Otimização: Gradiente usando Diferenças Finitas

W Atual	W + h (1ªdim)	Gradiente dW
[0.34, -1.11, 0.78, 0.12, 0.55, 2.81, -3.1, -1.5, 0.33,…]	[0.34 + **0.0001**, -1.11, 0.78, 0.12, 0.55, 2.81, -3.1, -1.5, 0.33,…]	[-2.5, ?, ?, ?, ?,…]

(1.25322 - 1.25347)/0.0001
= -2.5

$$\frac{df(x)}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Otimização: Gradiente usando Diferenças Finitas

W Atual	W + h (1ª dim)	Gradiente dW
[0.34,	[0.34,	[-2.5,
-1.11,	-1.11 + **0.0001**,	?,
0.78,	0.78,	?,
0.12,	0.12,	?,
0.55,	0.55,	?,
2.81,	2.81,	?,
-3.1,	-3.1,	?,
-1.5,	-1.5,	?,
0.33,…]	0.33,…]	?,…]

Otimização: Gradiente usando Diferenças Finitas

W Atual	W + h (1ª dim)	Gradiente dW
[0.34,	[0.34,	[-2.5,
-1.11,	-1.11 + **0.0001**,	**0.6**,
0.78,	0.78,	?,
0.12,	0.12,	?,
0.55,	0.55,	
2.81,	2.81,	
-3.1,	-3.1,	
-1.5,	-1.5,	
0.33,…]	0.33,…]	?,…]

(1.25353 - 1.25347)/0.0001
= 0.6

$$\frac{df(x)}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

O Custo é uma função de W

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

$$s = f(x; W) = Wx$$

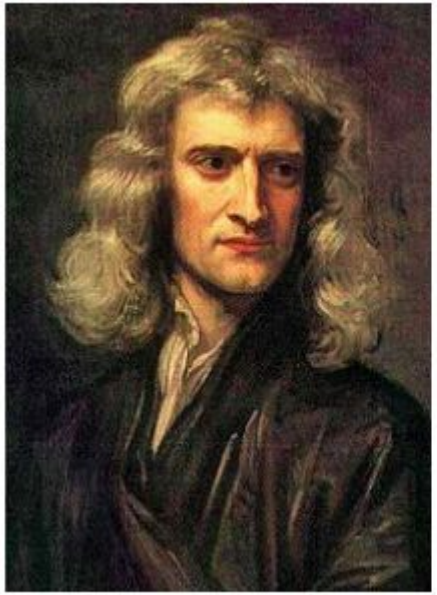
- Queremos $\nabla_W L$

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Esta imagem está em domínio público



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$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

$$s = f(x; W) = Wx$$

- Queremos $\nabla_W L$

Use cálculo para calcular o **gradiente analítico**.



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Otimização: Gradiente

W Atual

[0.34,
-1.11,
0.78,
0.12,
0.55,
2.81,
-3.1,
-1.5,
0.33,…]

Custo: 1.25347

dW = …
(alguma função
de W)

Gradiente dW

[-2.5,
0.6,
0,
0.2,
0.7,
-0.5,
1.1,
1.3,
-2.1,…]

Resumo

Gradiente numérico: Aproximado, lento, fácil de escrever

Gradiente analítico: Exato, rápido, sujeito a erros

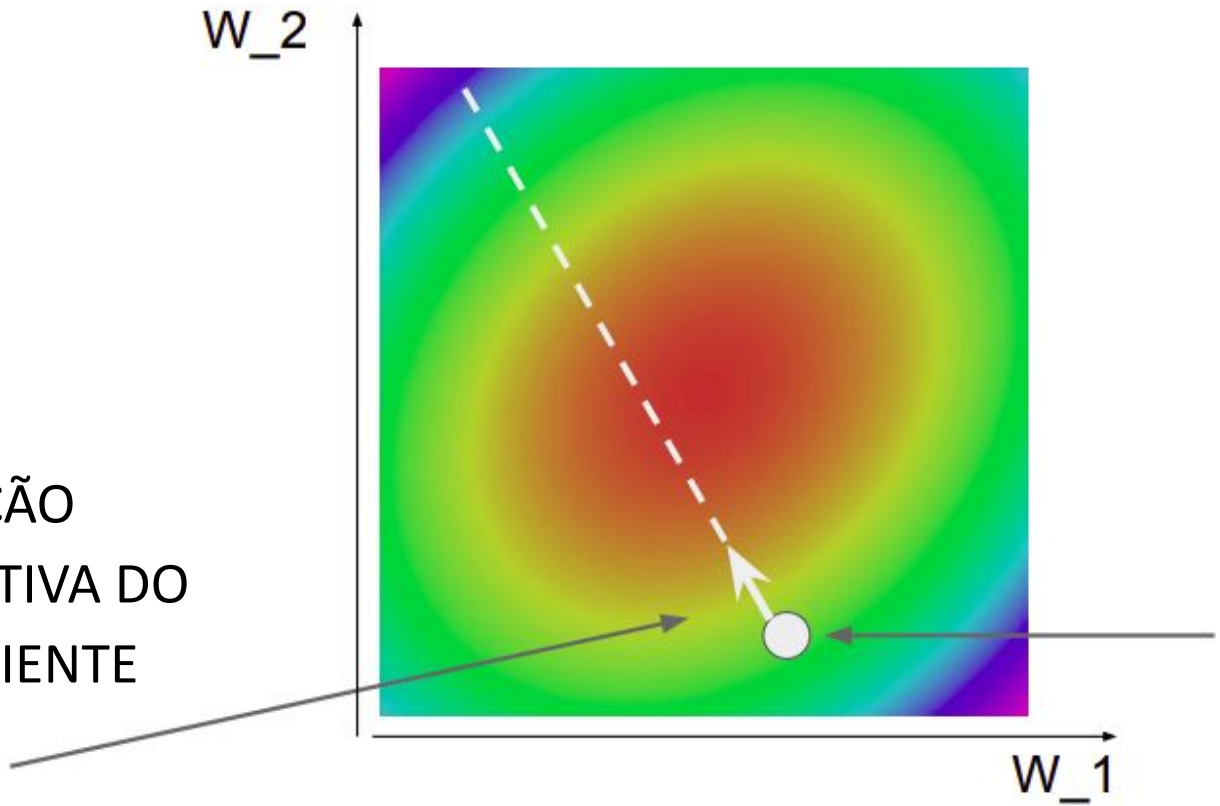
Na prática: sempre use o gradiente analítico, mas verifique a implementação usando o gradiente numérico. Isso é chamado de verificação de gradiente (gradient check).

Gradiente Descendente

```
# Vanilla Gradient Descent

while True:
  weights_grad = evaluate_gradient(loss_fun, data, weights)
  weights += - step_size * weights_grad # perform parameter update
```

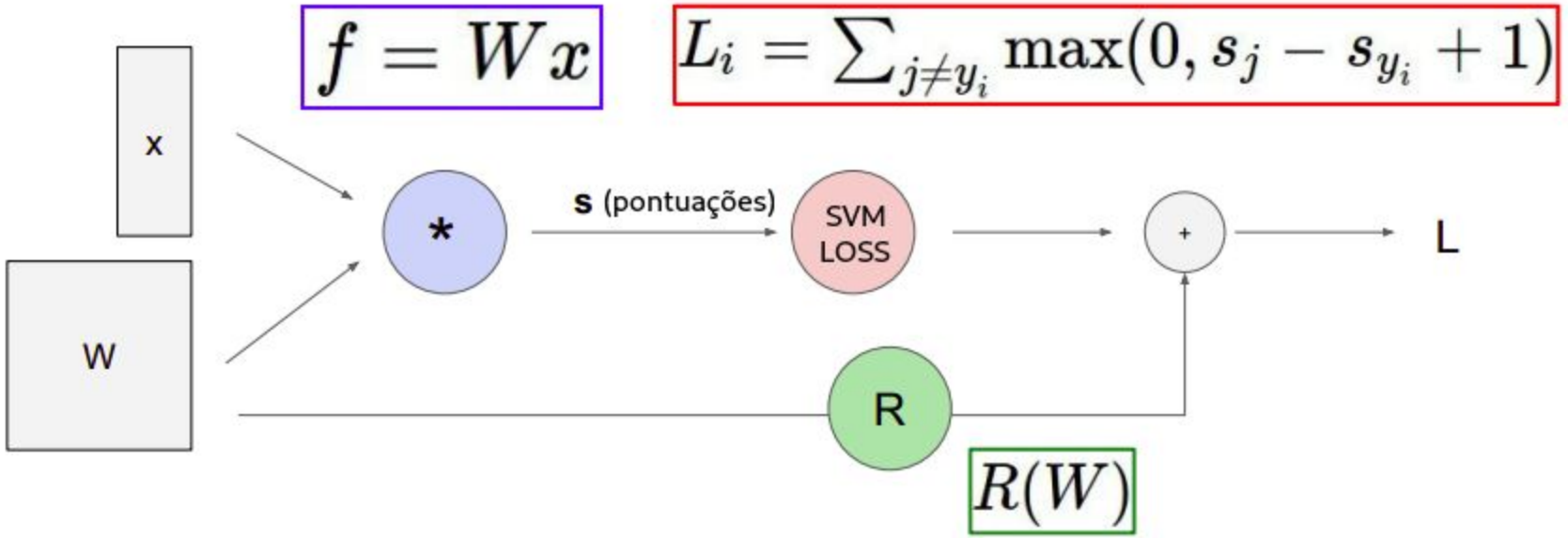
Otimização: Gradiente Descendente



DIREÇÃO NEGATIVA DO GRADIENTE

Backpropagation

Grafo Computacional

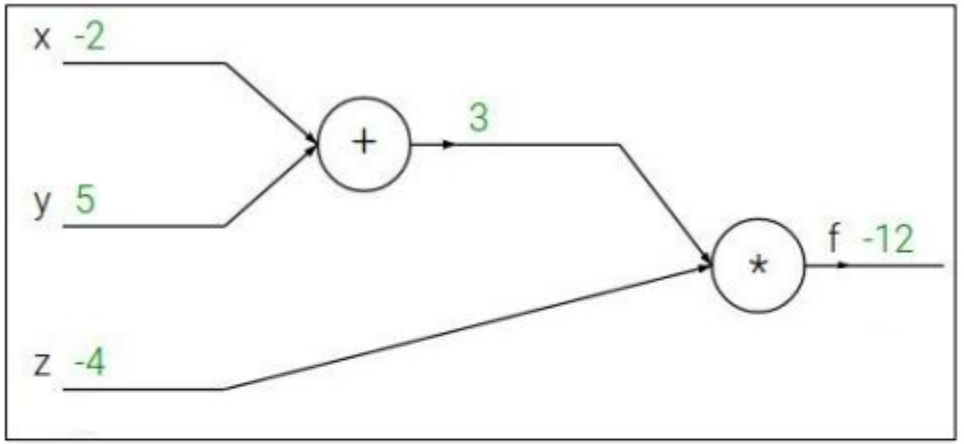


Um exemplo simples de Backpropagation

Backpropagation: Um exemplo simples

$$f(x, y, z) = (x + y)z$$

e.g. x = -2, y = 5, z = -4



Um exemplo simples de Backpropagation

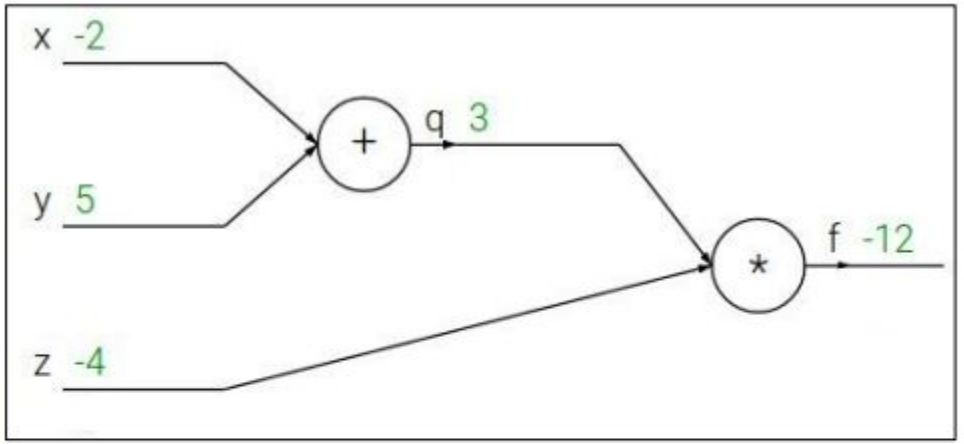
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Um exemplo simples de Backpropagation

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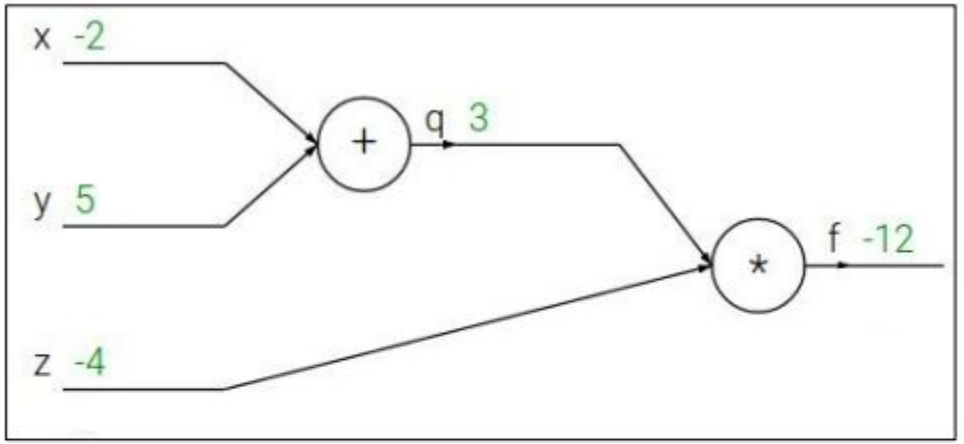
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Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



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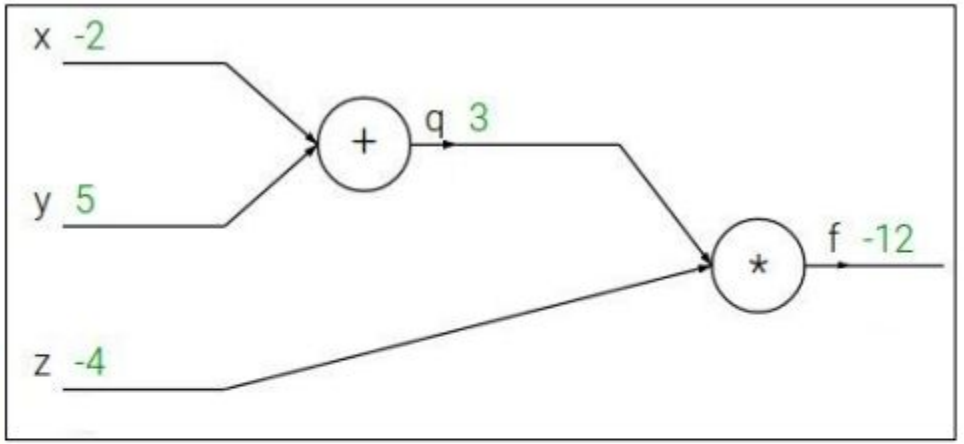
$$f(x, y, z) = (x + y)z$$

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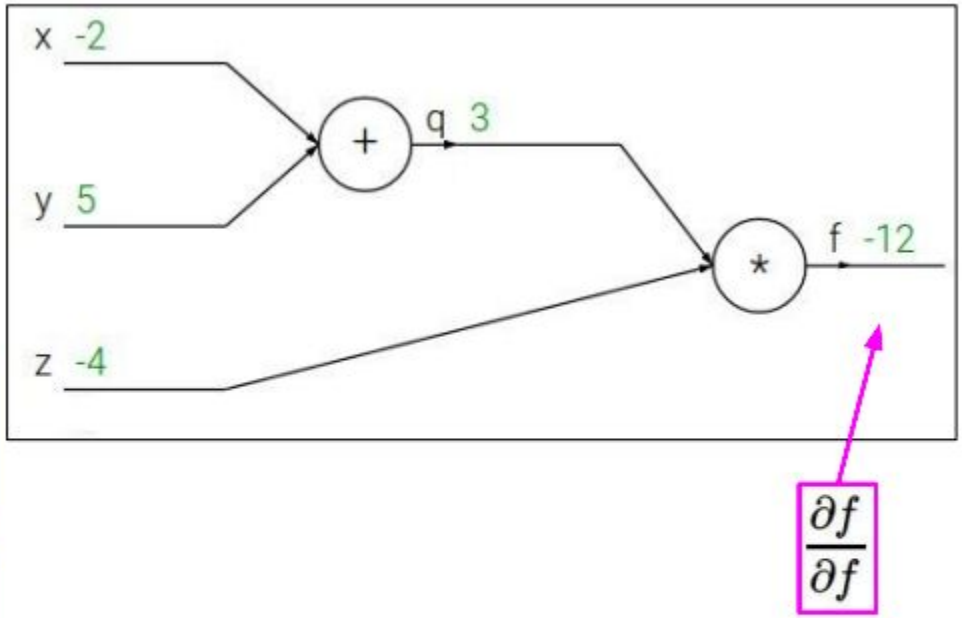
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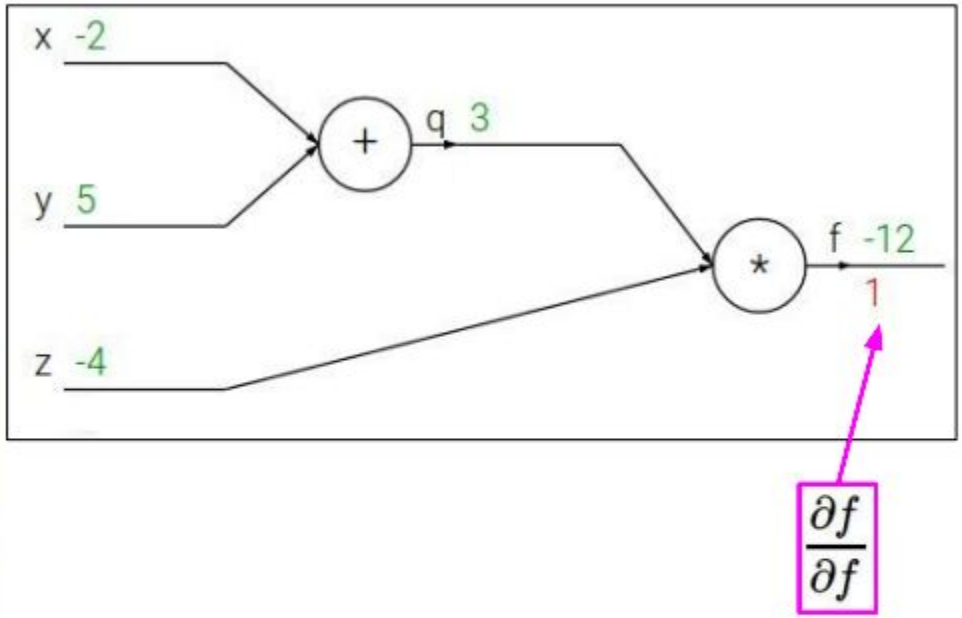
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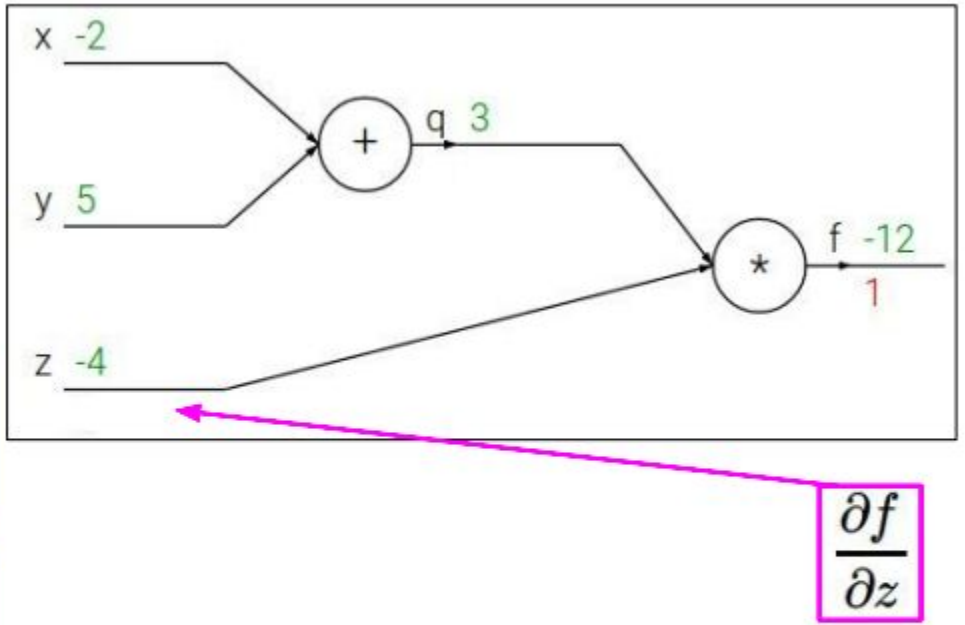
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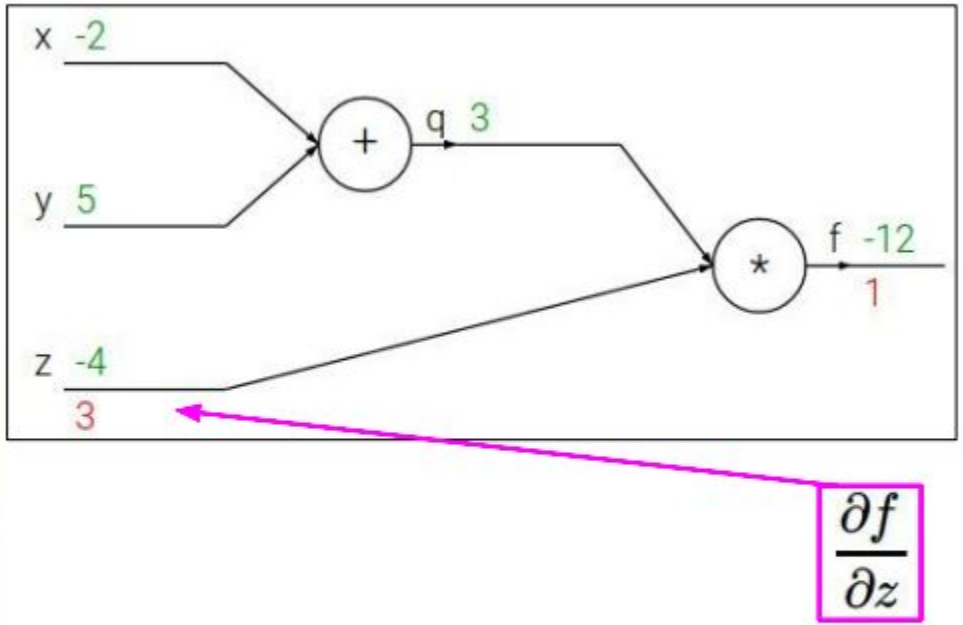
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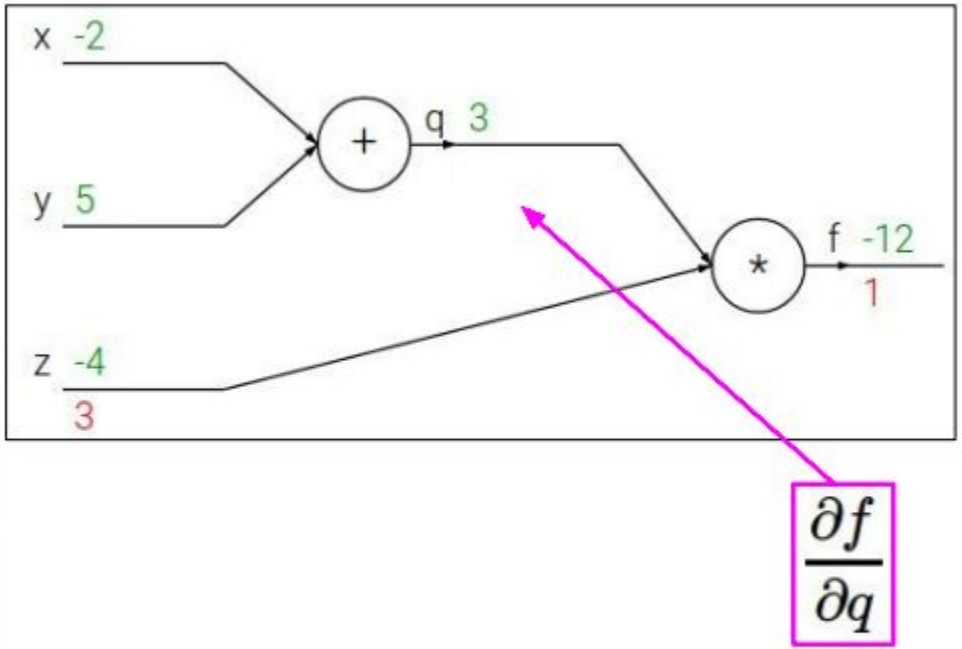
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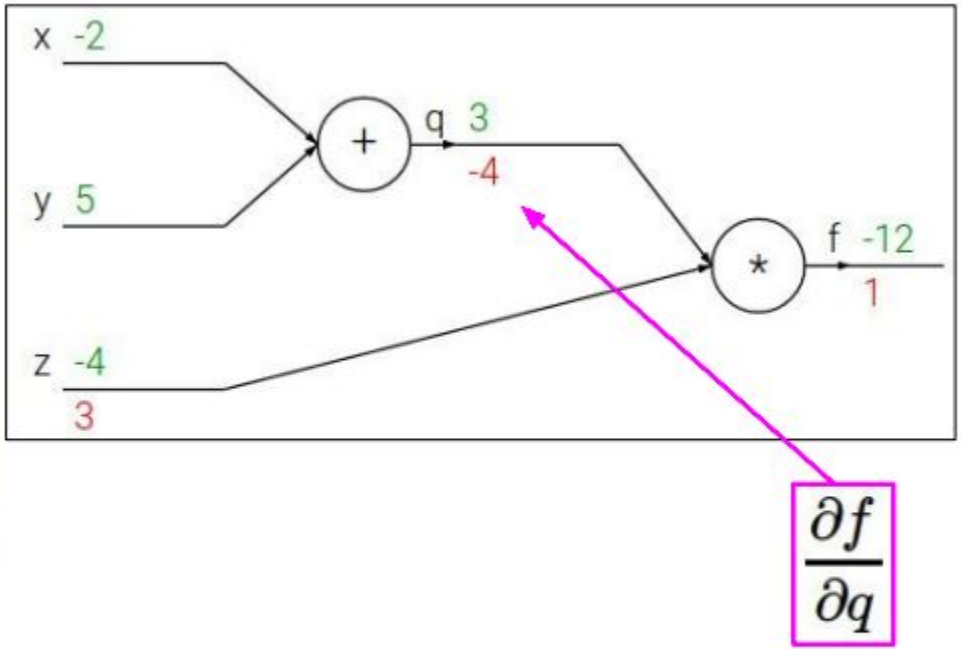
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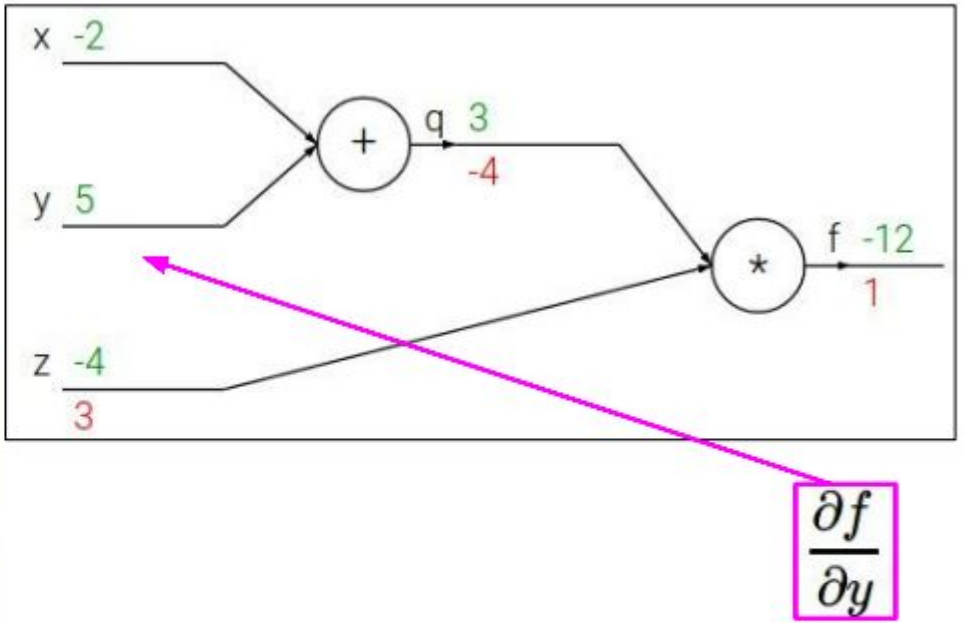
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Um exemplo simples de Backpropagation

Backpropagation: Um exemplo simples

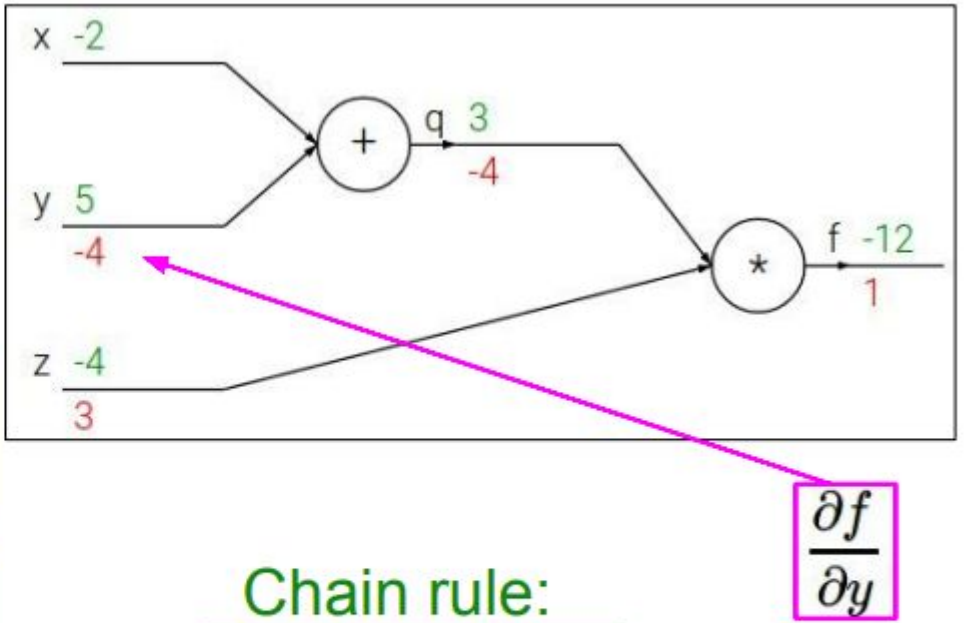
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Chain rule:

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q}\frac{\partial q}{\partial y}$$

Um exemplo simples de Backpropagation

Backpropagation: Um exemplo simples

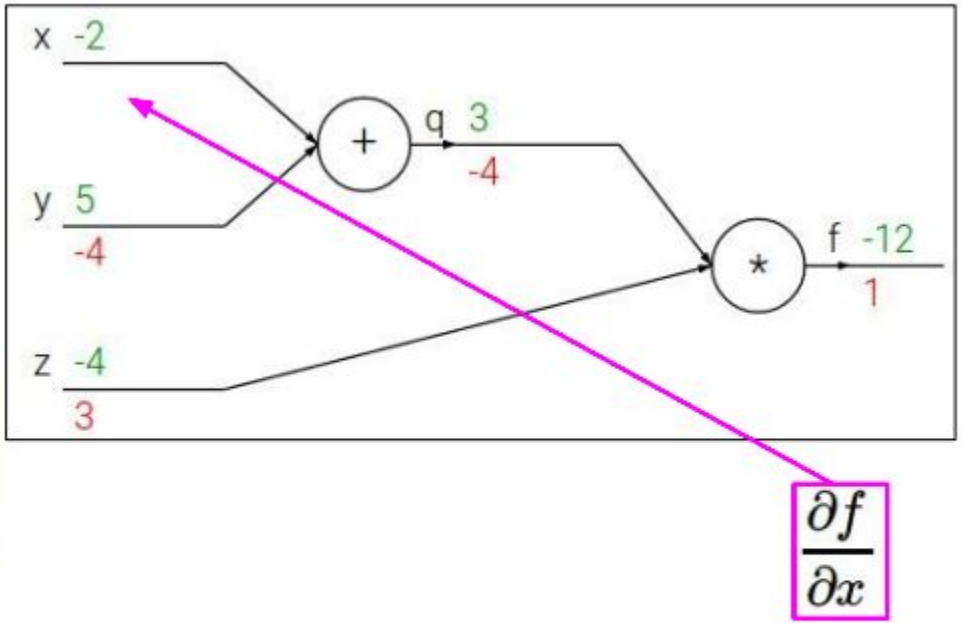
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Um exemplo simples de Backpropagation

Backpropagation: Um exemplo simples

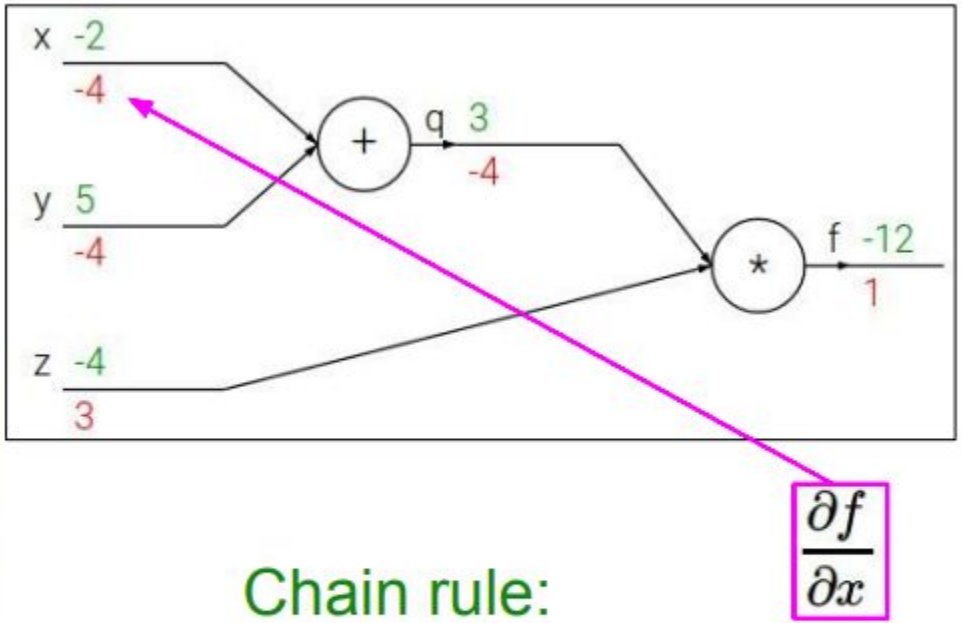
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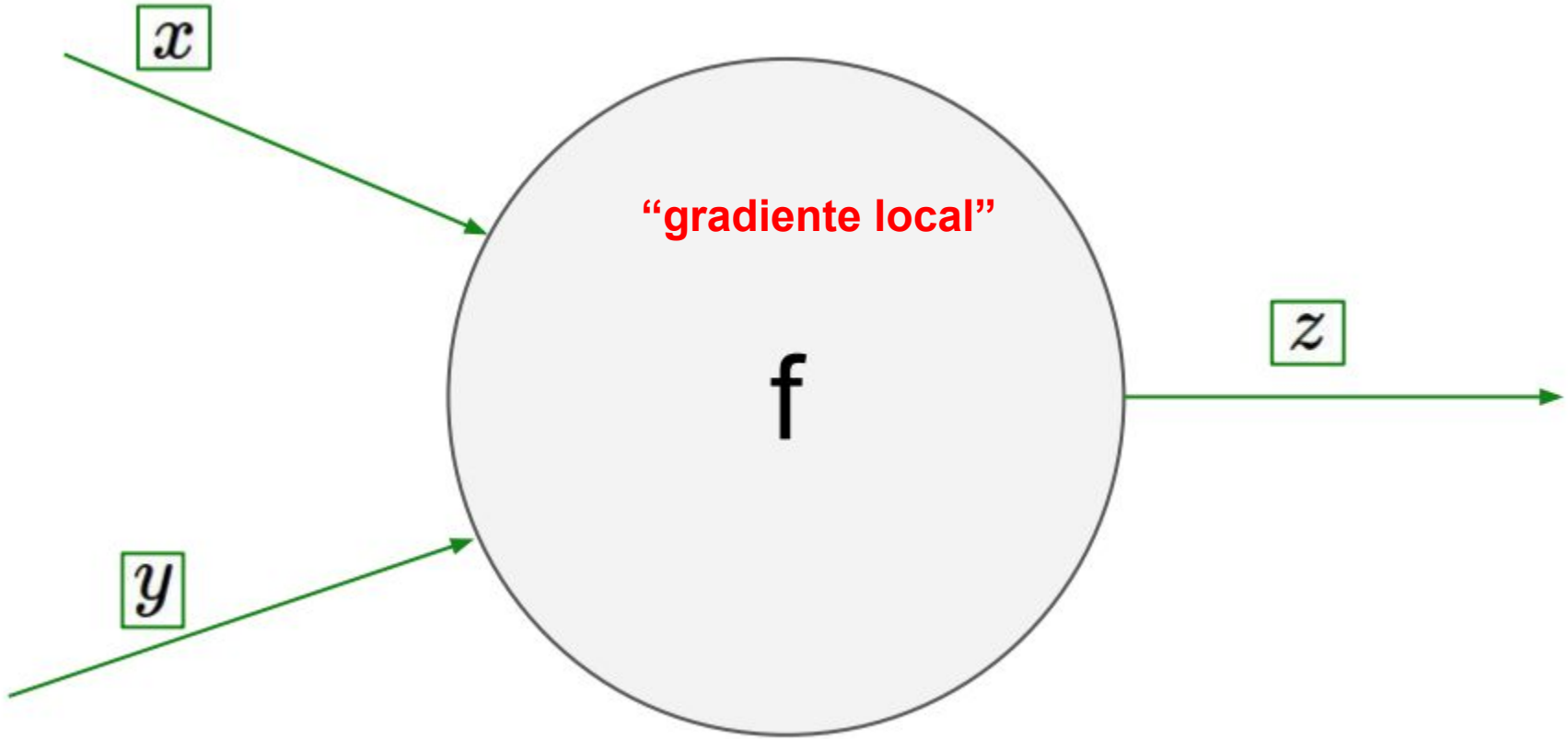
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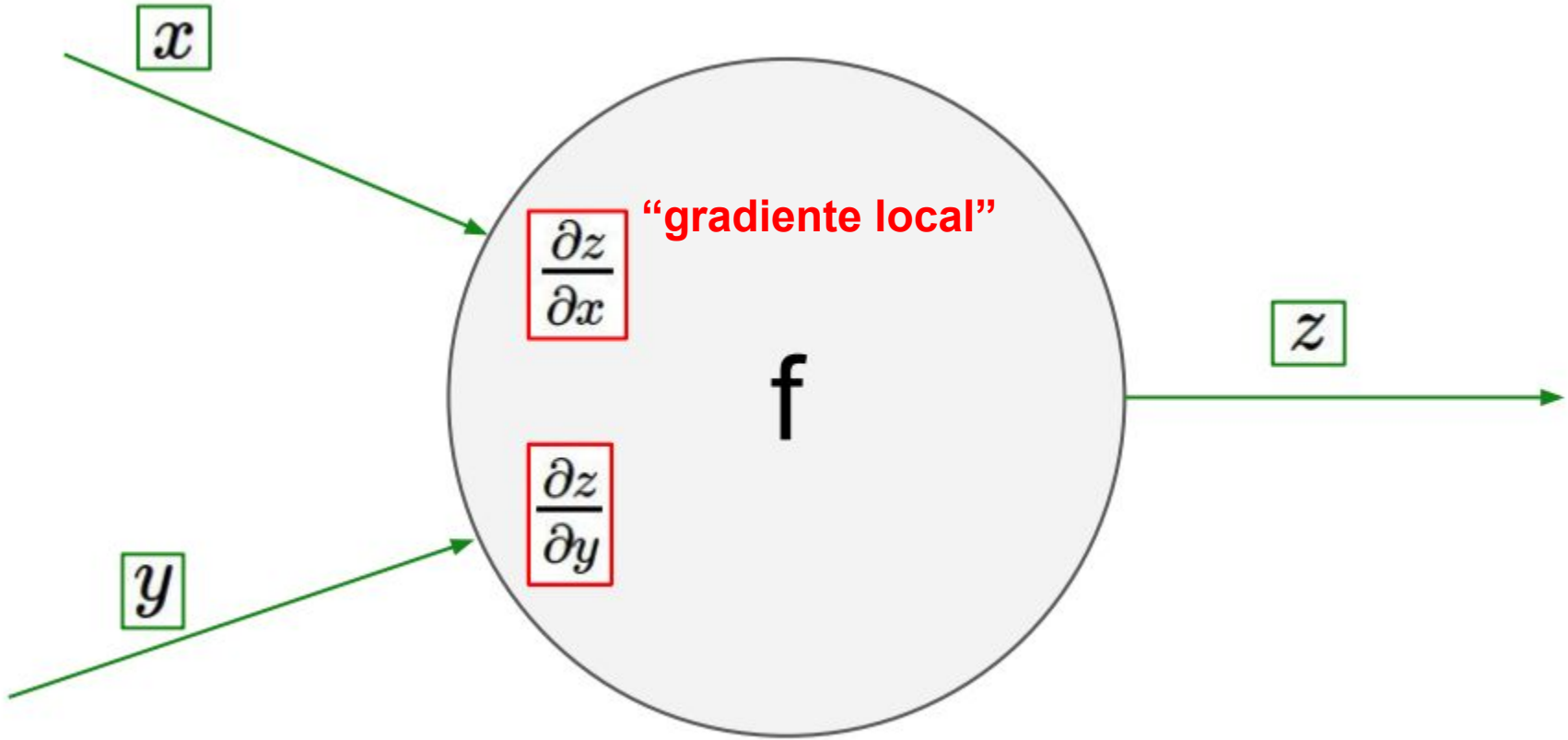
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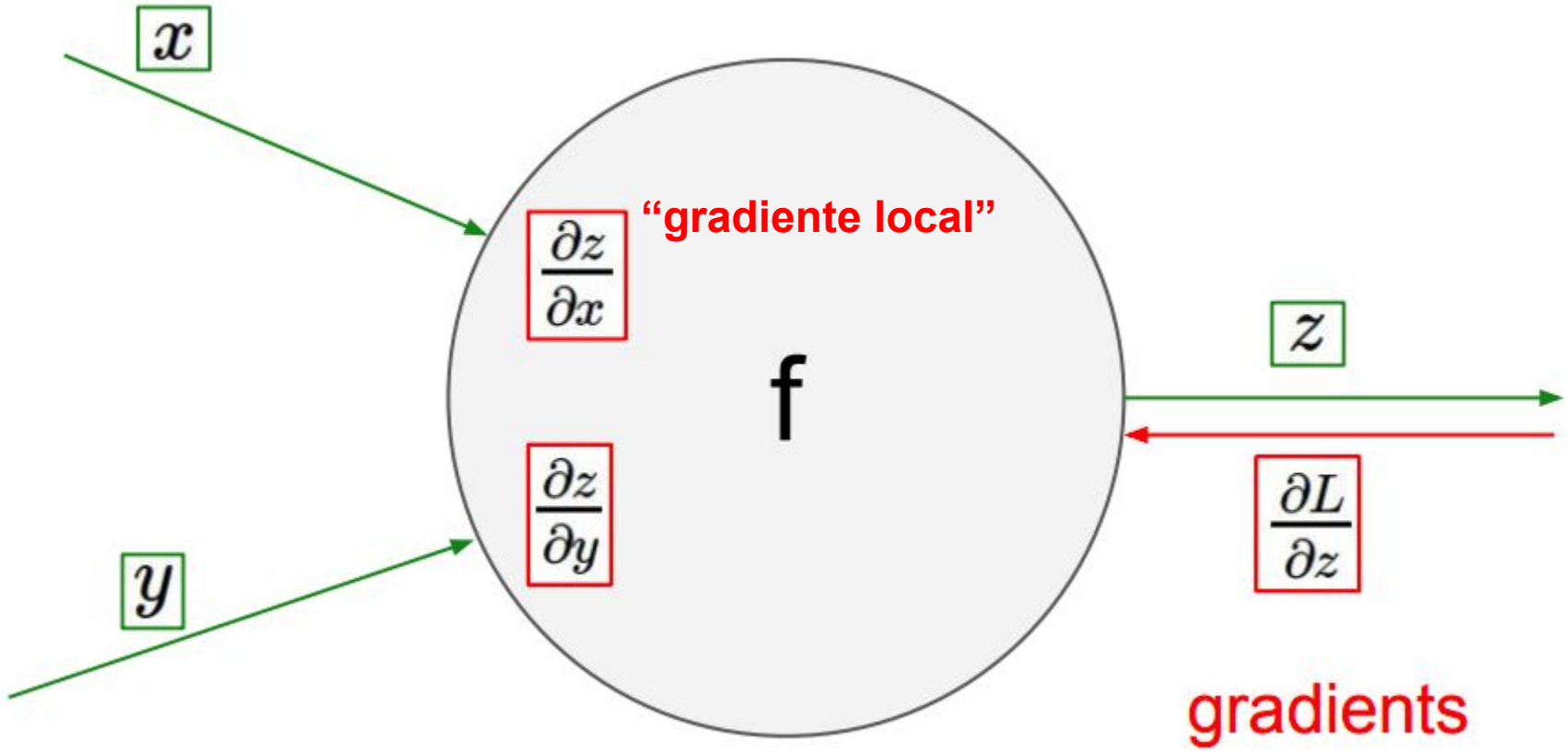
Um exemplo simples de Backpropagation



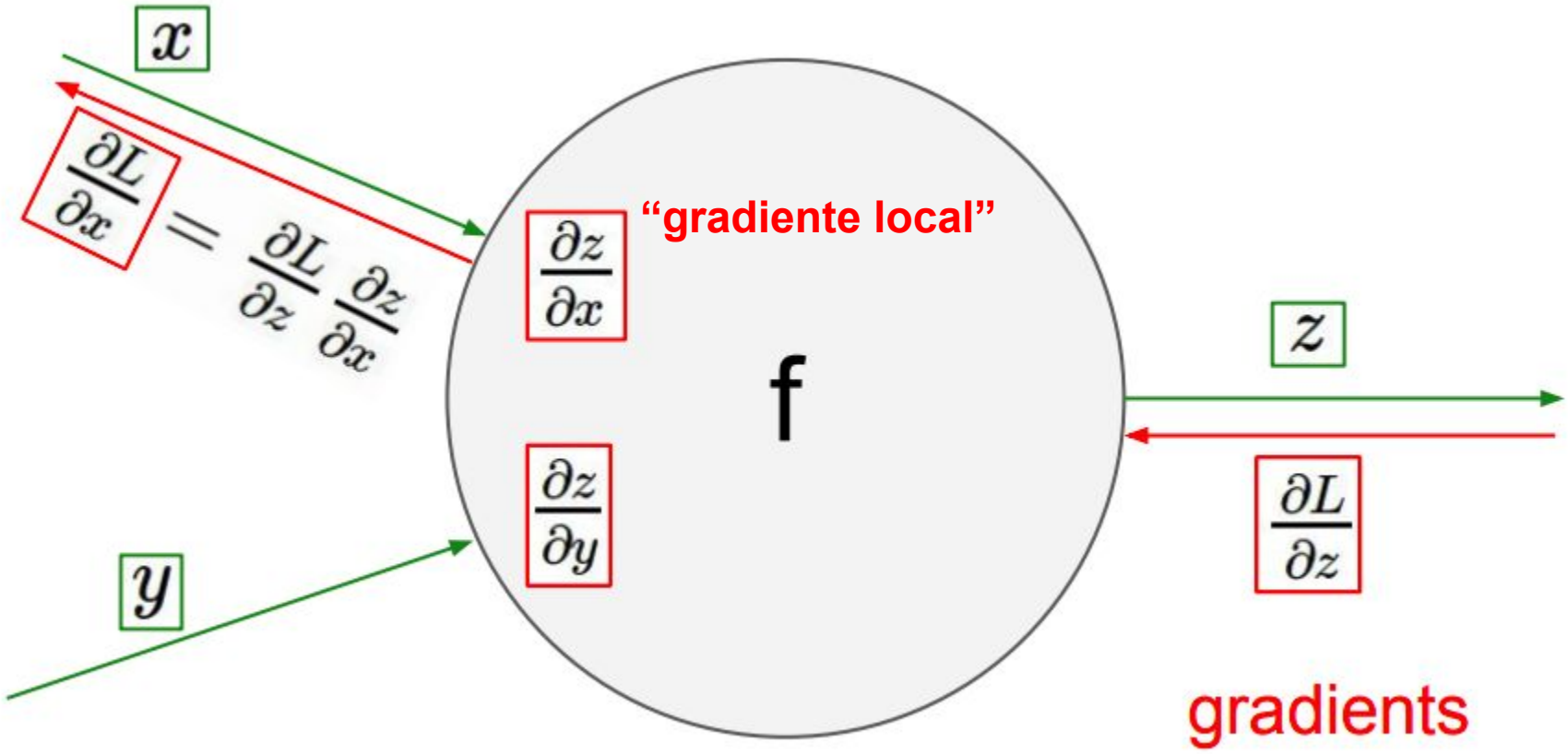
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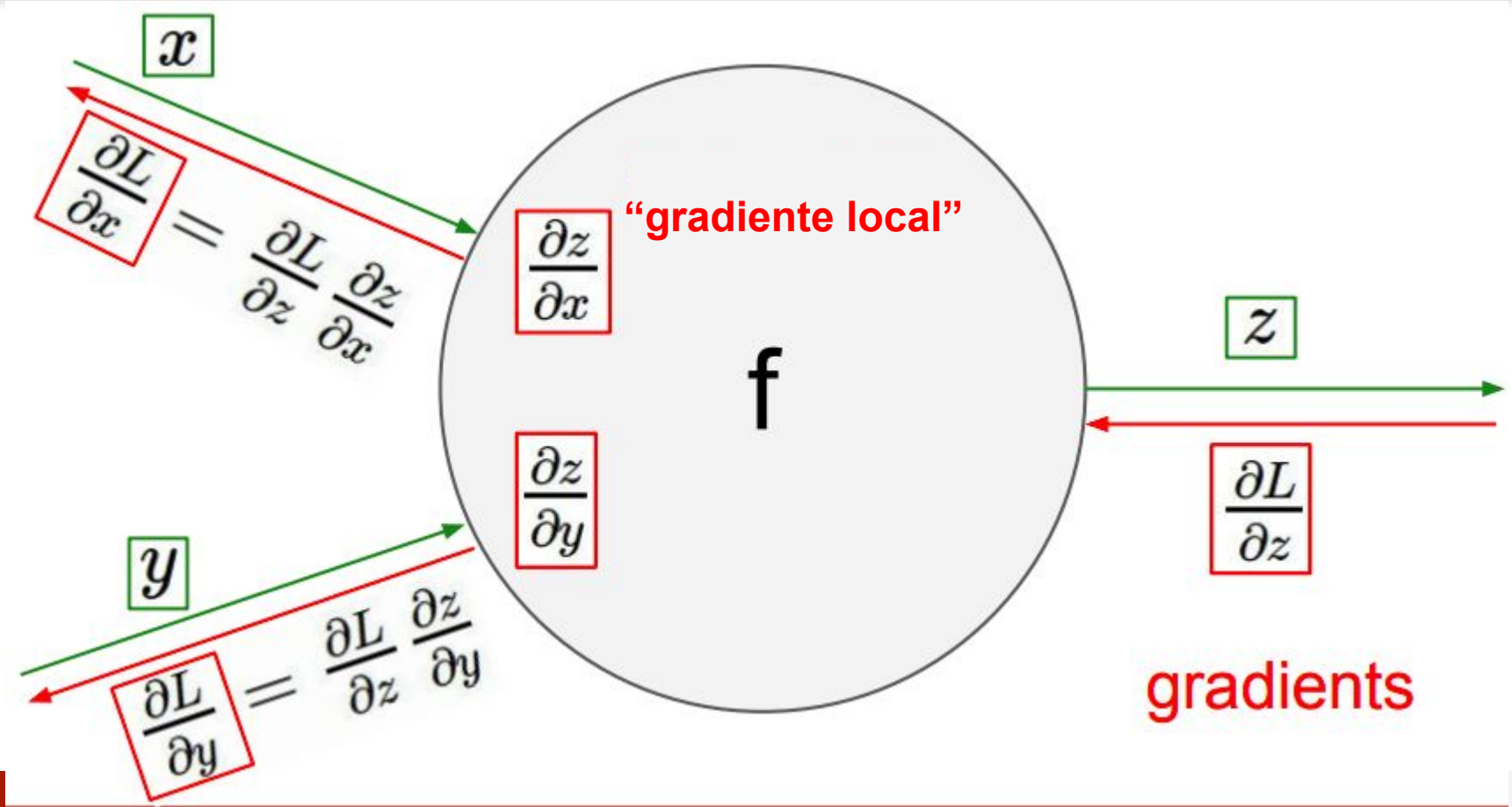
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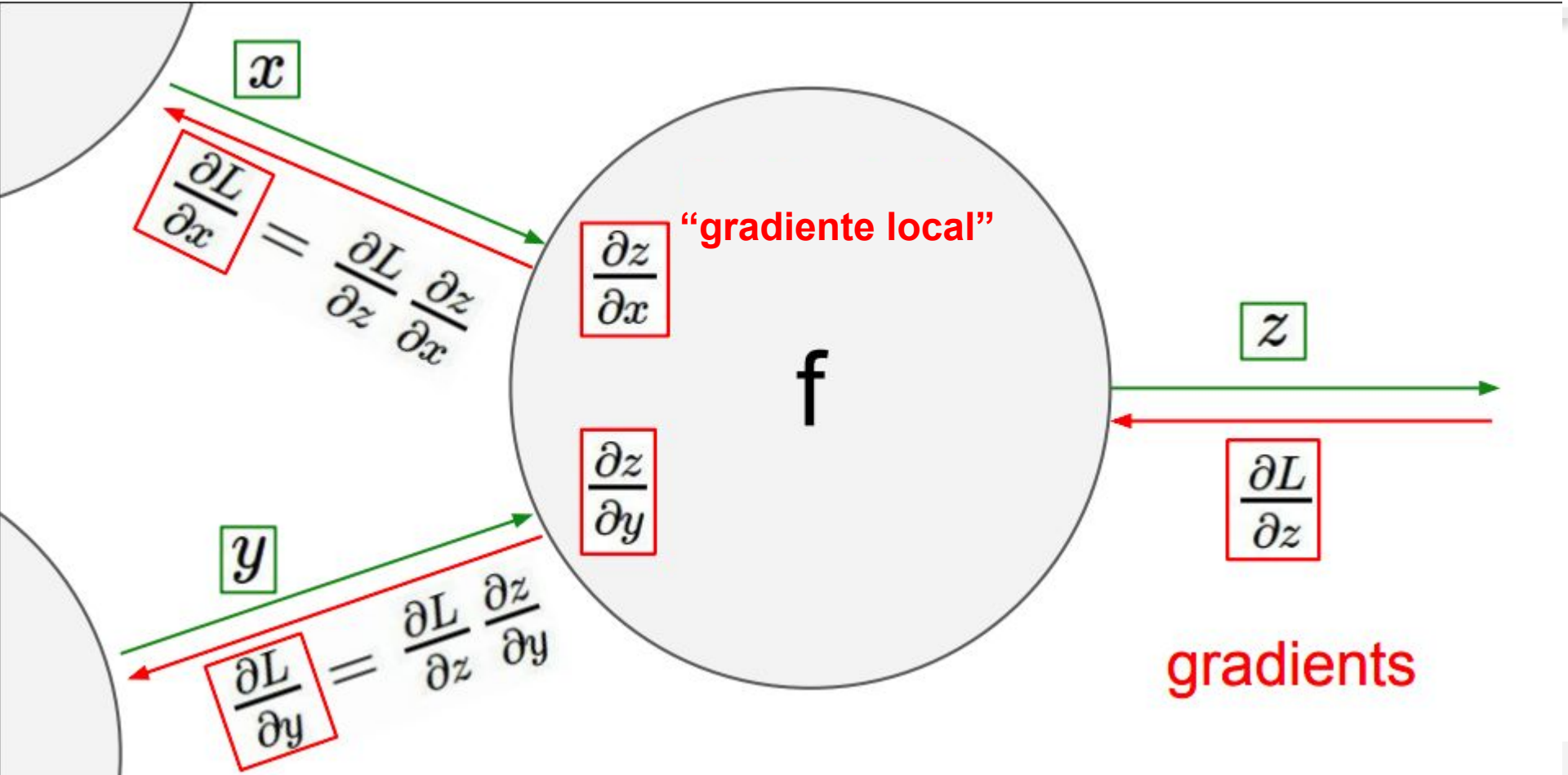
Um exemplo simples de Backpropagation



Um exemplo simples de Backpropagation



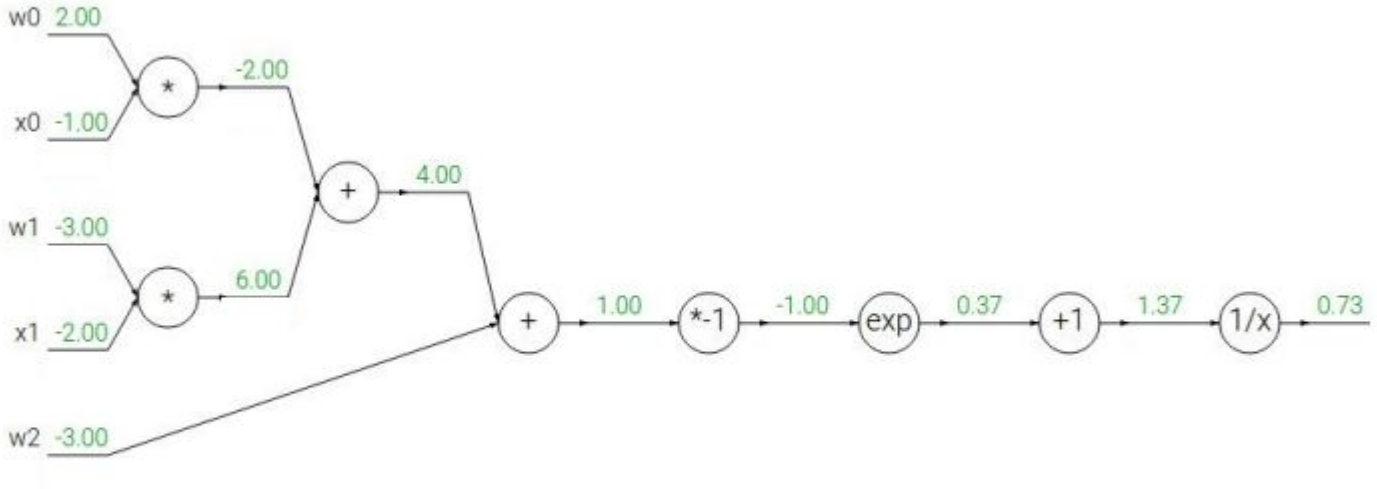
Um exemplo simples de Backpropagation



Outro Exemplo

Outro exemplo

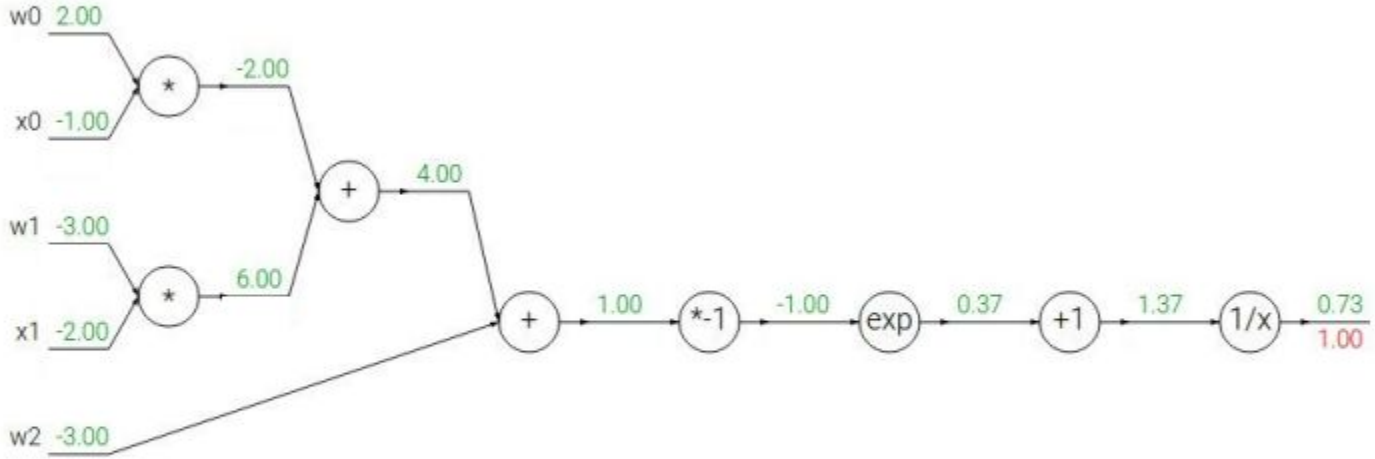
$$f(w,x) = \frac{1}{1+e^{-(w_0x_0+w_1x_1+w_2)}}$$



Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

$$f_a(x) = ax \quad \rightarrow \quad \frac{df}{dx} = a$$

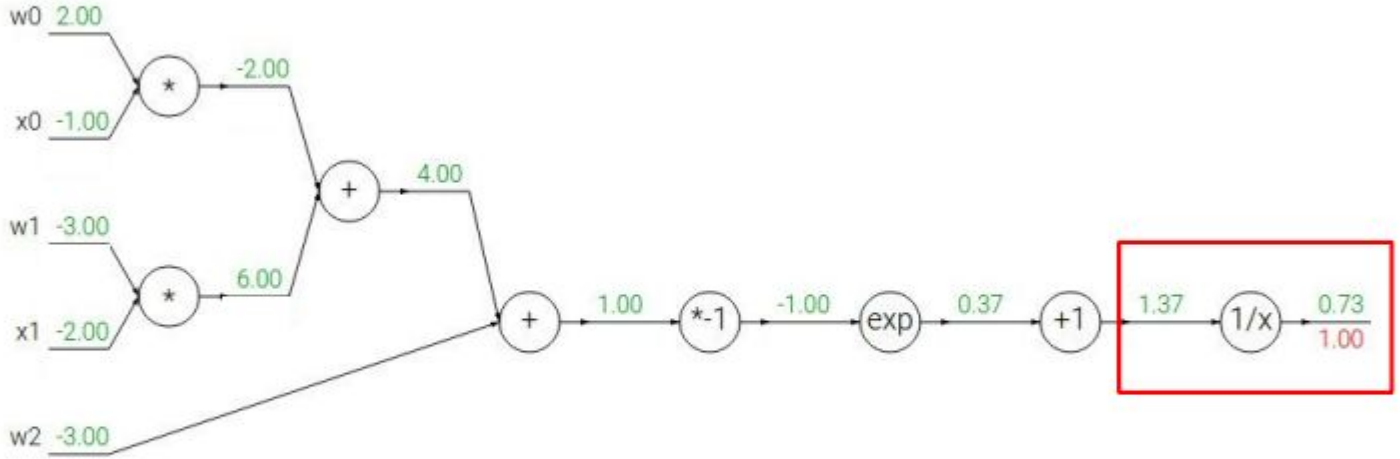
$$f(x) = \frac{1}{x} \quad \rightarrow \quad \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \quad \rightarrow \quad \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

$$f_a(x) = ax \quad \rightarrow \quad \frac{df}{dx} = a$$

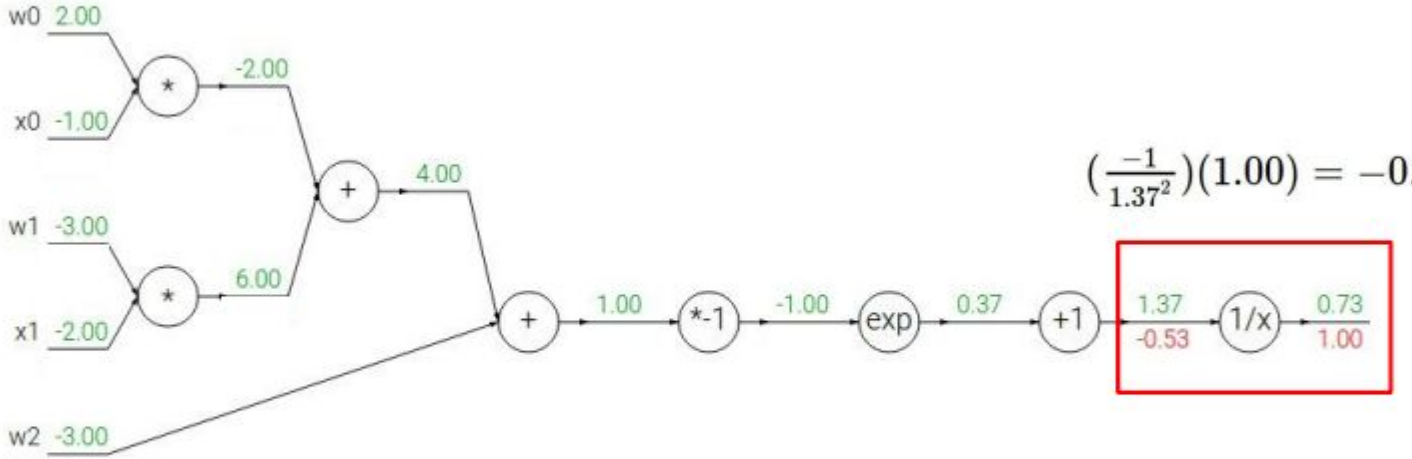
$$f(x) = \frac{1}{x} \quad \rightarrow \quad \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \quad \rightarrow \quad \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1+e^{-(w_0x_0+w_1x_1+w_2)}}$$



$$(\frac{-1}{1.37^2})(1.00) = -0.53$$

$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

$$f_a(x) = ax \quad \rightarrow \quad \frac{df}{dx} = a$$

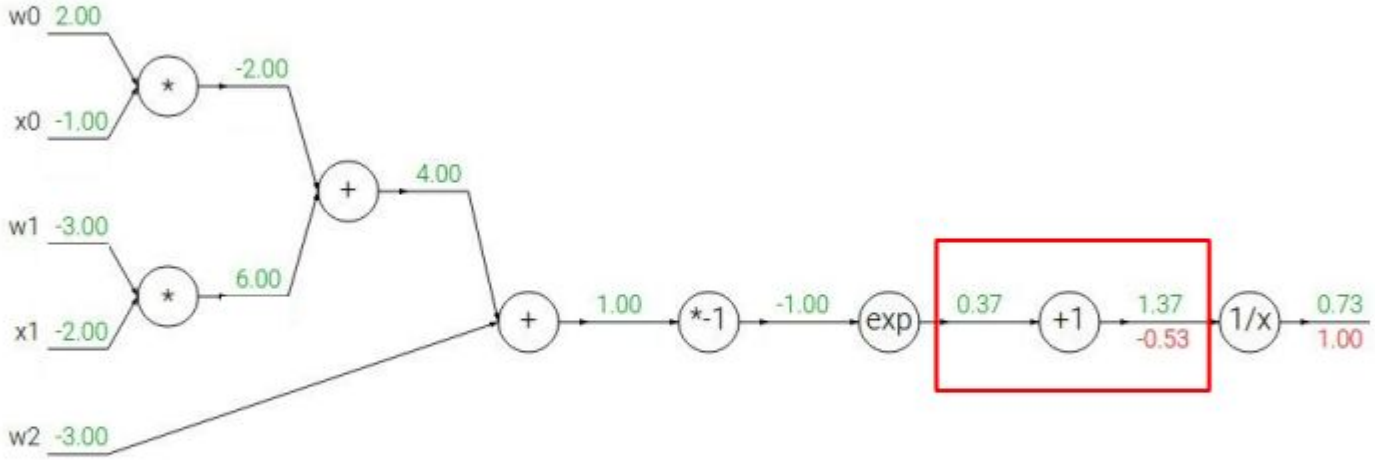
$$f(x) = \frac{1}{x} \quad \rightarrow \quad \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \quad \rightarrow \quad \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

$$f_a(x) = ax \quad \rightarrow \quad \frac{df}{dx} = a$$

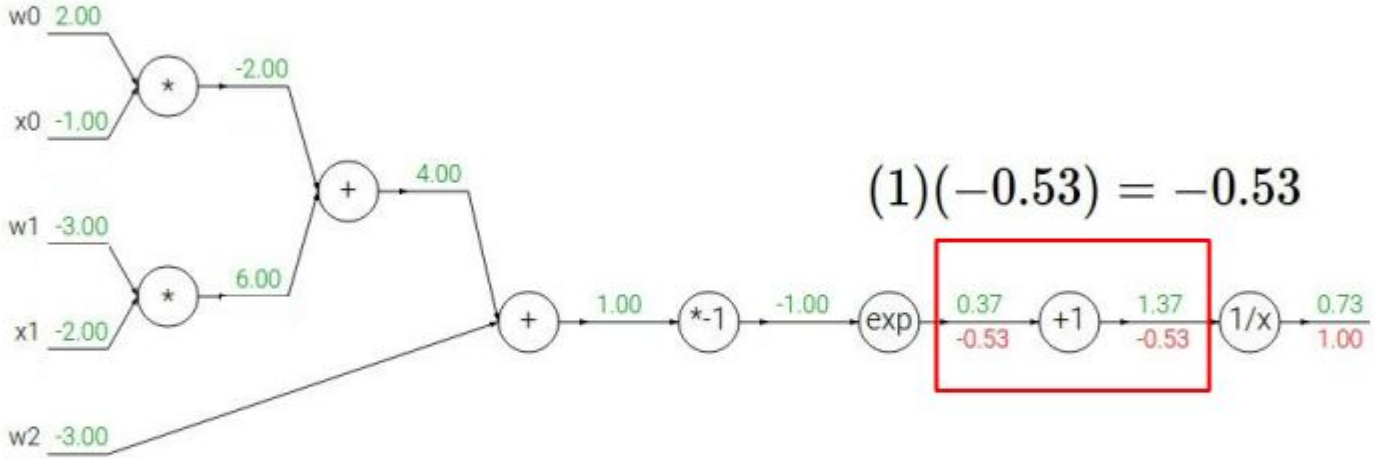
$$f(x) = \frac{1}{x} \quad \rightarrow \quad \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \quad \rightarrow \quad \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$(1)(-0.53) = -0.53$$

$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

$$f_a(x) = ax \quad \rightarrow \quad \frac{df}{dx} = a$$

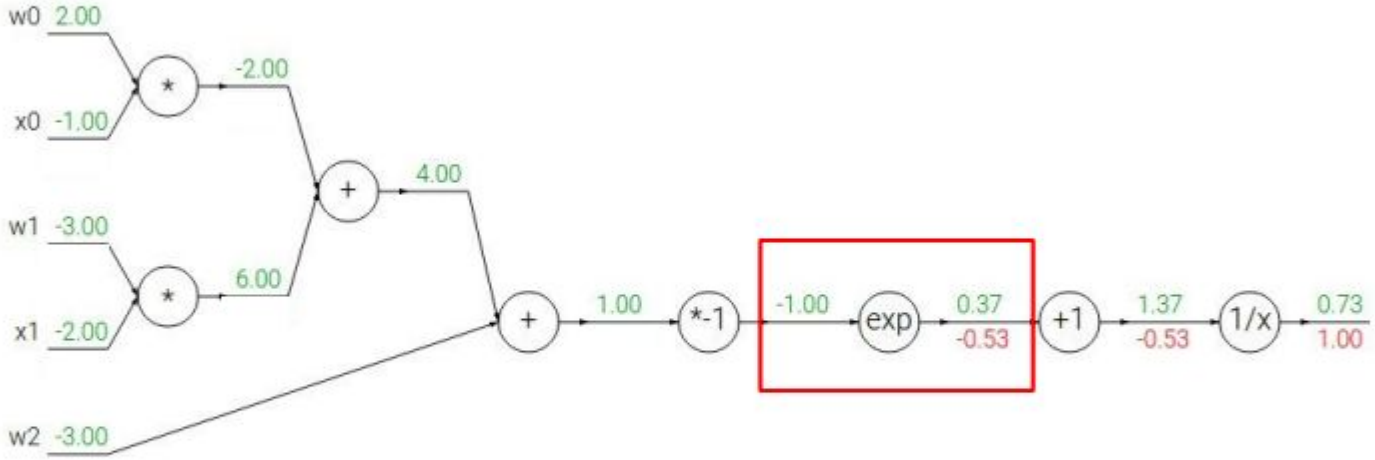
$$f(x) = \frac{1}{x} \quad \rightarrow \quad \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \quad \rightarrow \quad \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

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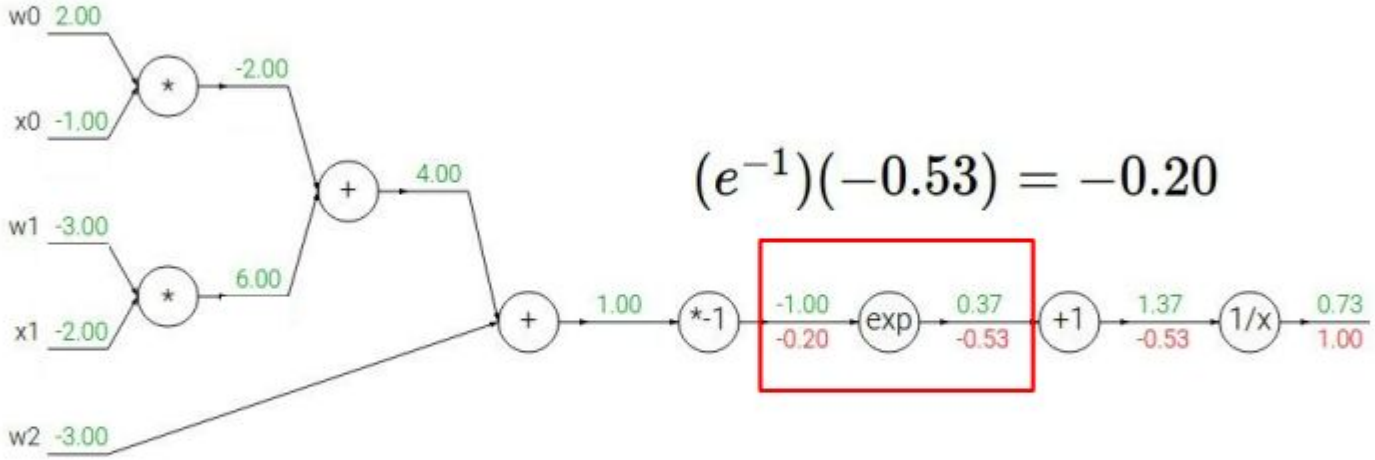
$$f(x) = \frac{1}{x} \quad \rightarrow \quad \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \quad \rightarrow \quad \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1+e^{-(w_0x_0+w_1x_1+w_2)}}$$



$$(e^{-1})(-0.53) = -0.20$$

$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

$$f_a(x) = ax \quad \rightarrow \quad \frac{df}{dx} = a$$

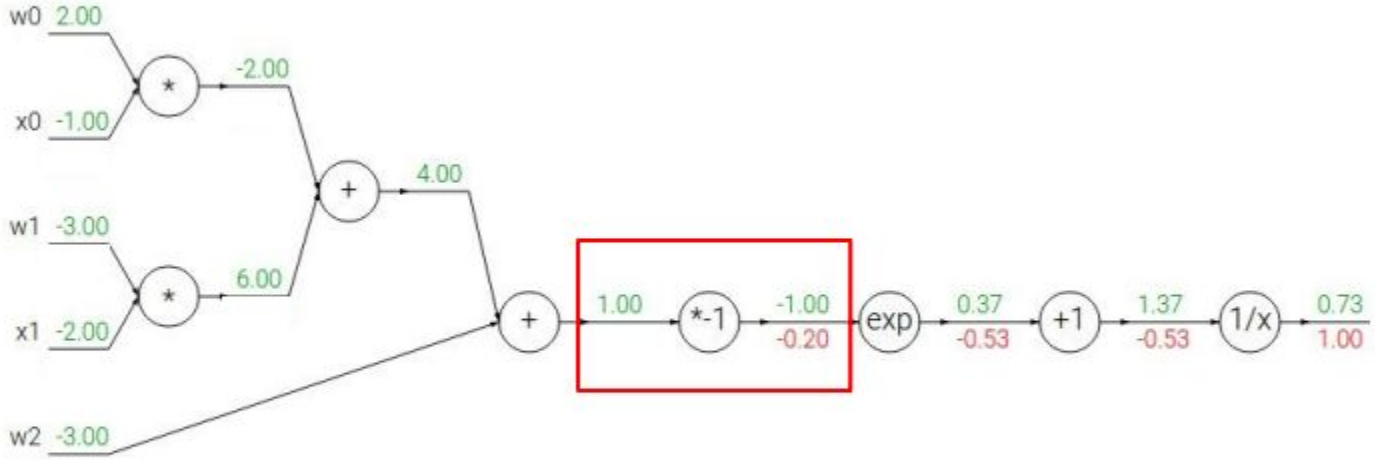
$$f(x) = \frac{1}{x} \quad \rightarrow \quad \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \quad \rightarrow \quad \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

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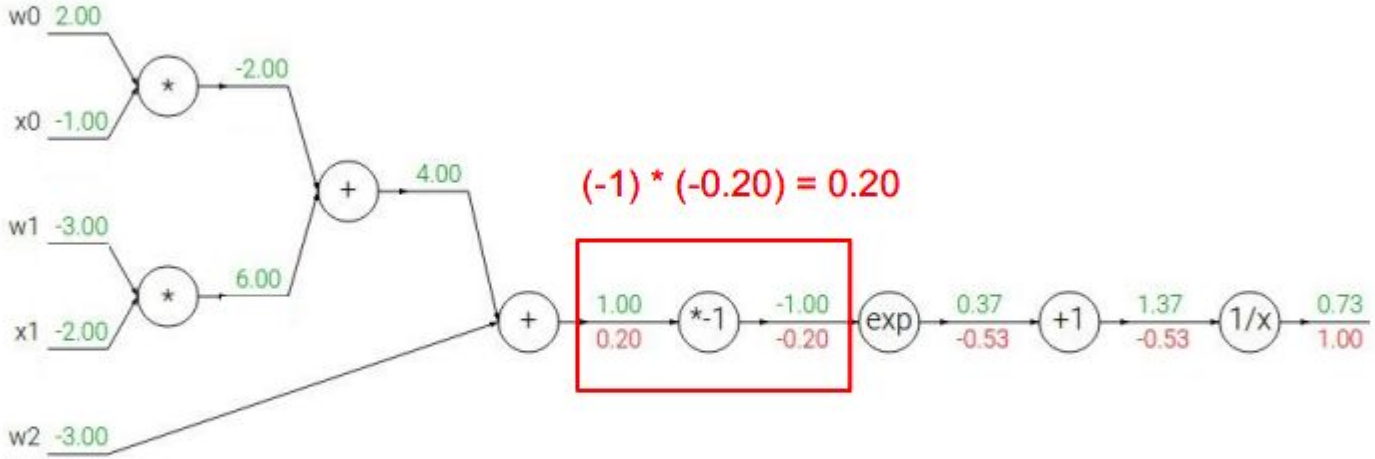
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Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

$$f_a(x) = ax \quad \rightarrow \quad \frac{df}{dx} = a$$

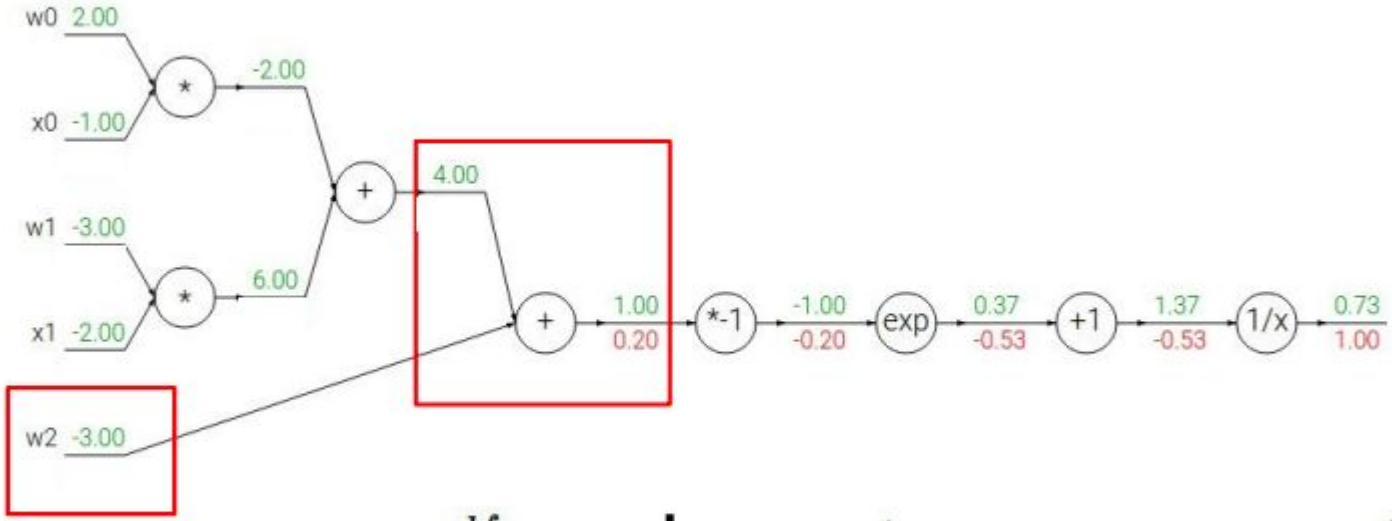
$$f(x) = \frac{1}{x} \quad \rightarrow \quad \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \quad \rightarrow \quad \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

$$f_a(x) = ax \rightarrow \frac{df}{dx} = a$$

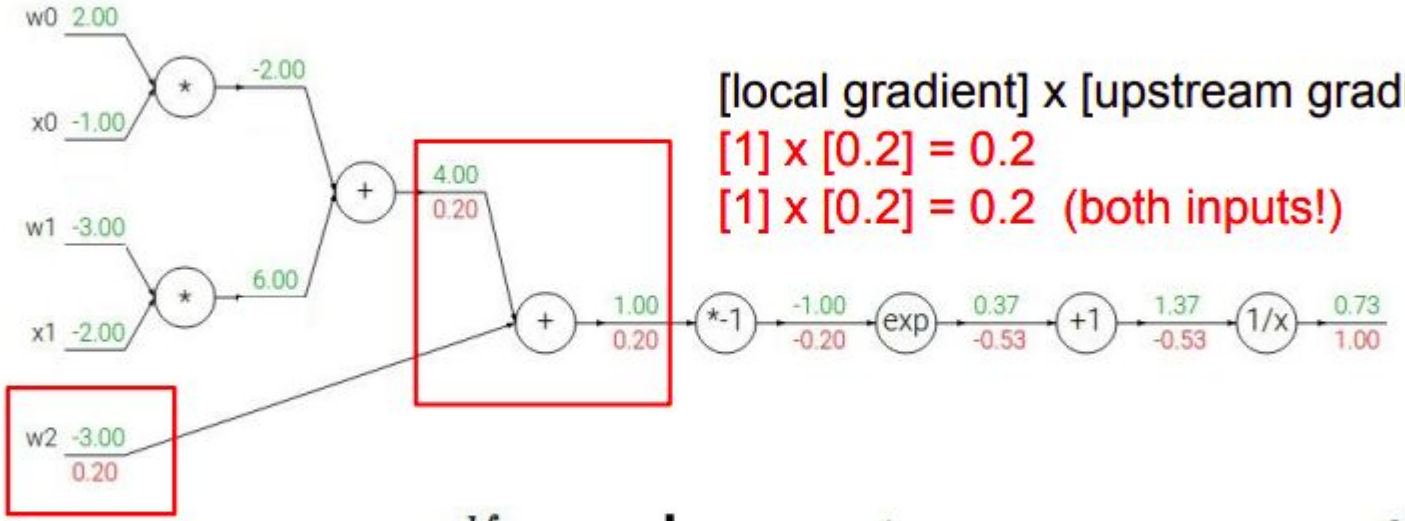
$$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



[local gradient] x [upstream gradient]

[1] x [0.2] = 0.2

[1] x [0.2] = 0.2 (both inputs!)

$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

$$f_a(x) = ax \rightarrow \frac{df}{dx} = a$$

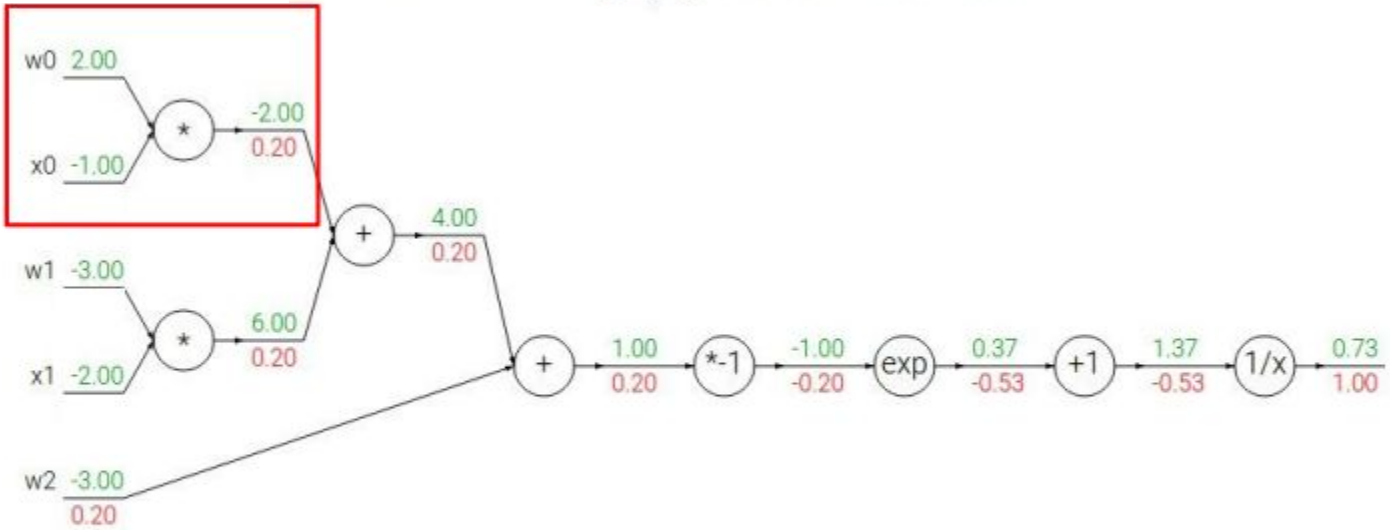
$$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$f(x) = e^x \quad \rightarrow \quad \frac{df}{dx} = e^x$$

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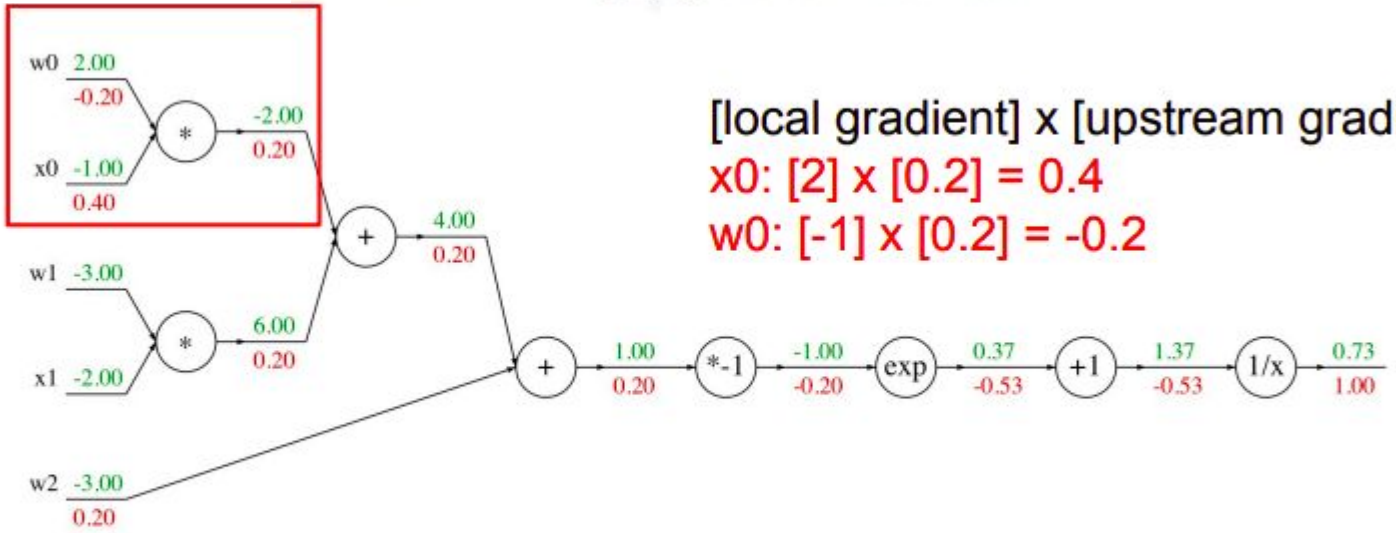
$$f(x) = \frac{1}{x} \quad \rightarrow \quad \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \quad \rightarrow \quad \frac{df}{dx} = 1$$

Outro Exemplo

Outro exemplo

$$f(w,x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



[local gradient] x [upstream gradient]

x0: [2] x [0.2] = 0.4

w0: [-1] x [0.2] = -0.2

$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

$$f_a(x) = ax \rightarrow \frac{df}{dx} = a$$

$$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$$

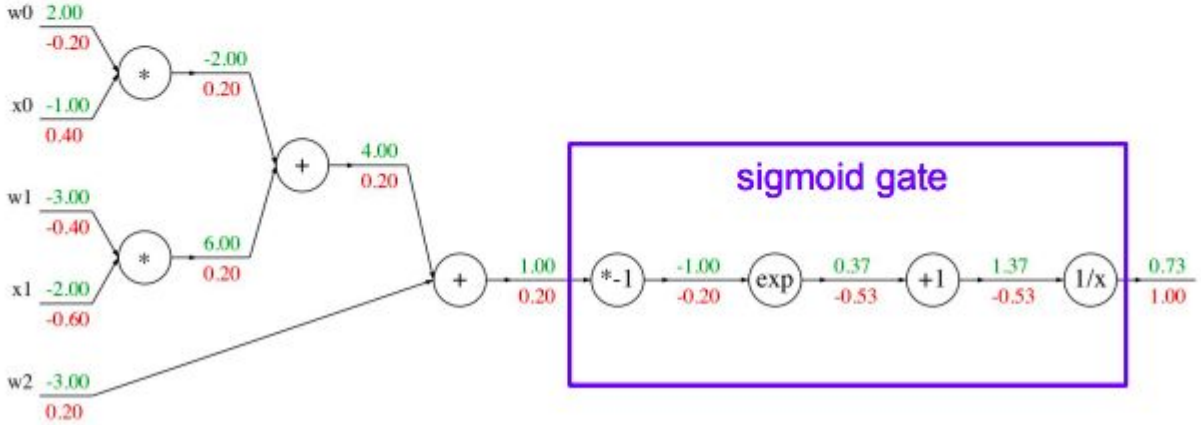
$$f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$$

Outro Exemplo

$$f(w,x) = \frac{1}{1+e^{-(w_0x_0+w_1x_1+w_2)}}$$

$$\boxed{\sigma(x) = \frac{1}{1+e^{-x}}}$$ sigmoid function

$$\frac{d\sigma(x)}{dx} = \frac{e^{-x}}{(1+e^{-x})^2} = \left(\frac{1+e^{-x}-1}{1+e^{-x}}\right)\left(\frac{1}{1+e^{-x}}\right) = (1-\sigma(x))\,\sigma(x)$$

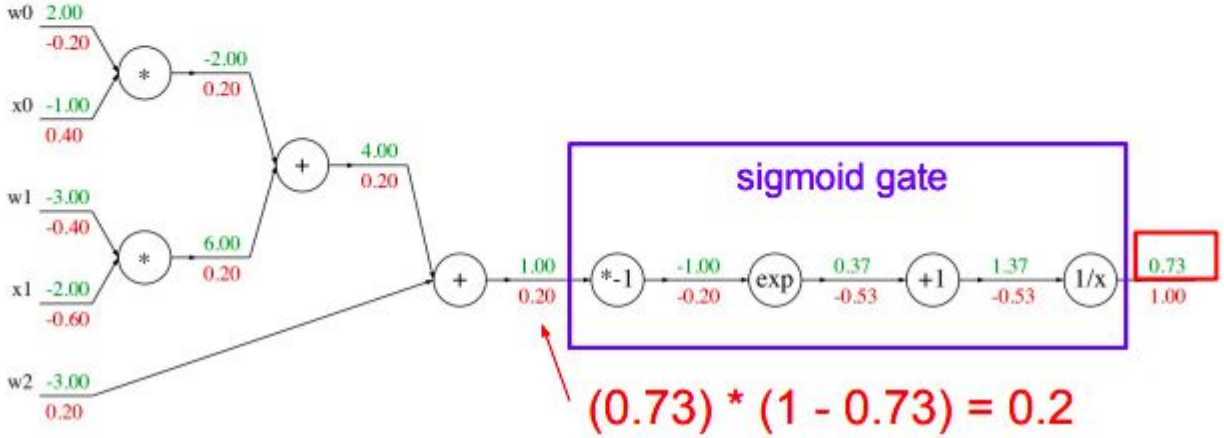


Outro Exemplo

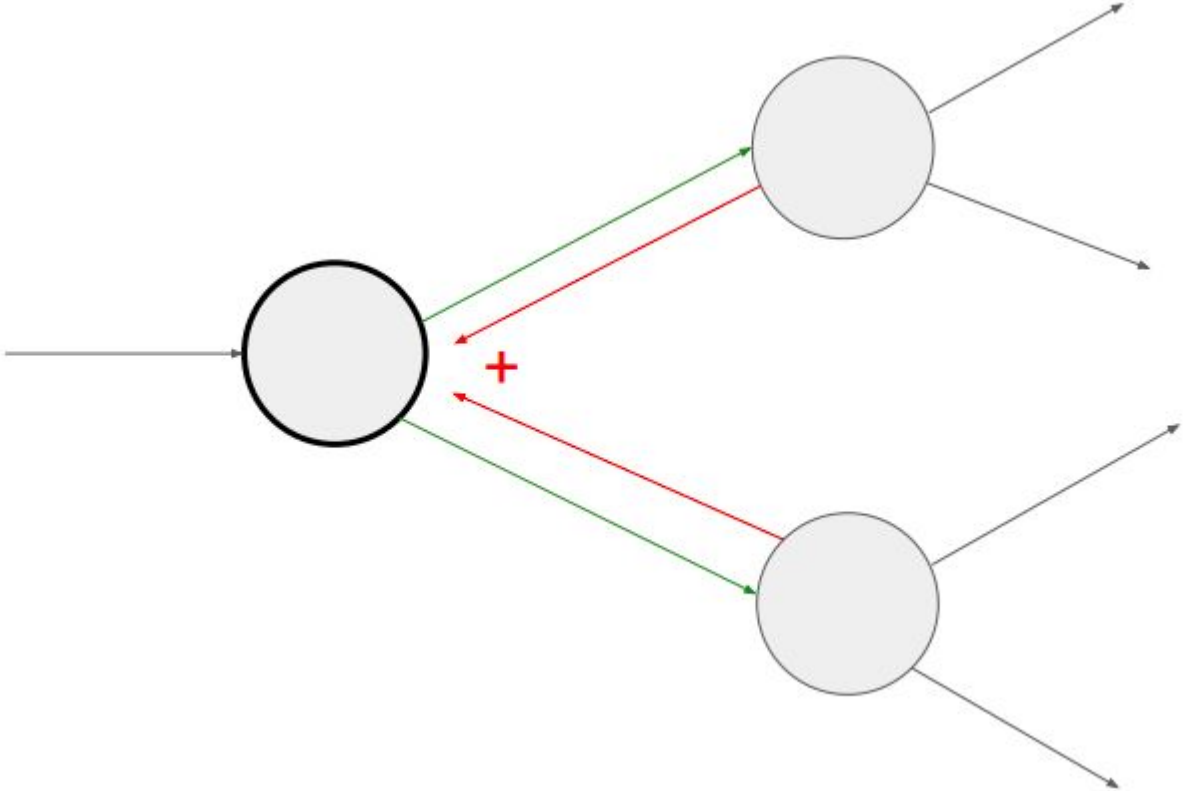
$$f(w,x) = \frac{1}{1+e^{-(w_0x_0+w_1x_1+w_2)}}$$

$$\boxed{\sigma(x) = \frac{1}{1+e^{-x}}}$$ sigmoid function

$$\frac{d\sigma(x)}{dx} = \frac{e^{-x}}{(1+e^{-x})^2} = \left(\frac{1+e^{-x}-1}{1+e^{-x}}\right)\left(\frac{1}{1+e^{-x}}\right) = (1-\sigma(x))\,\sigma(x)$$



Adição de gradiente nas ramificações



Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

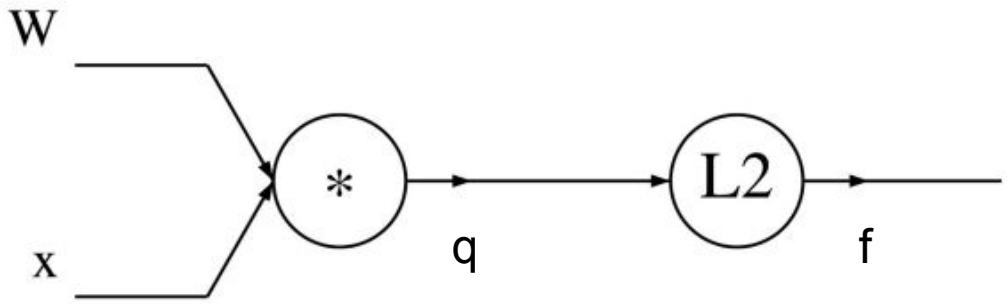
Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

$\in \mathbb{R}^n$ $\in \mathbb{R}^{n \times n}$

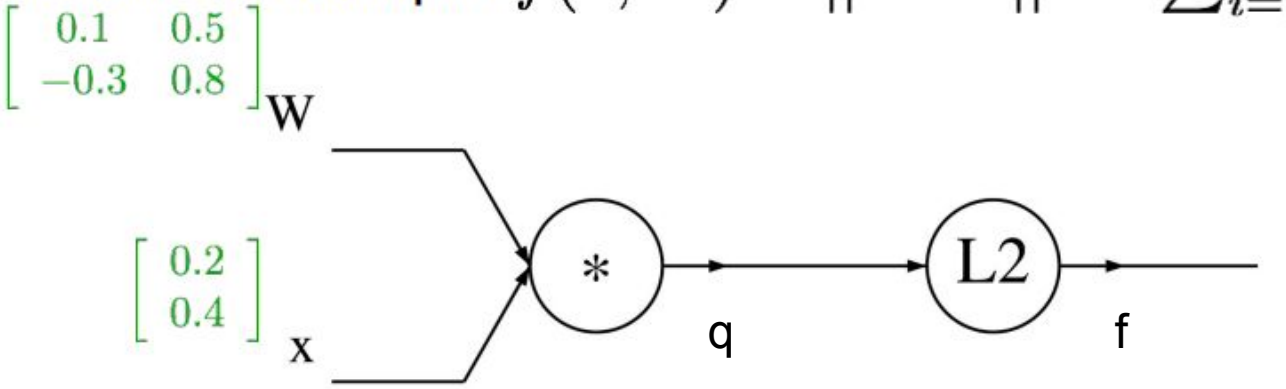
Um exemplo vetorizado

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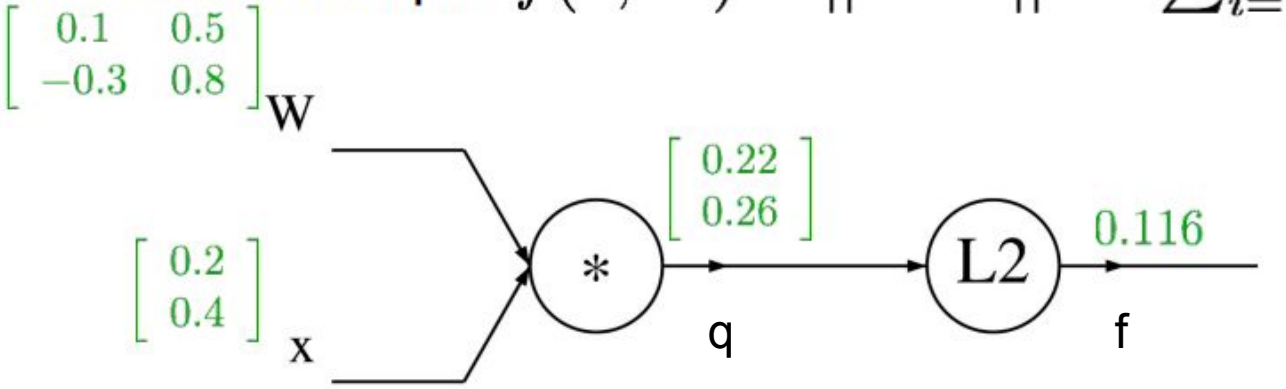
Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



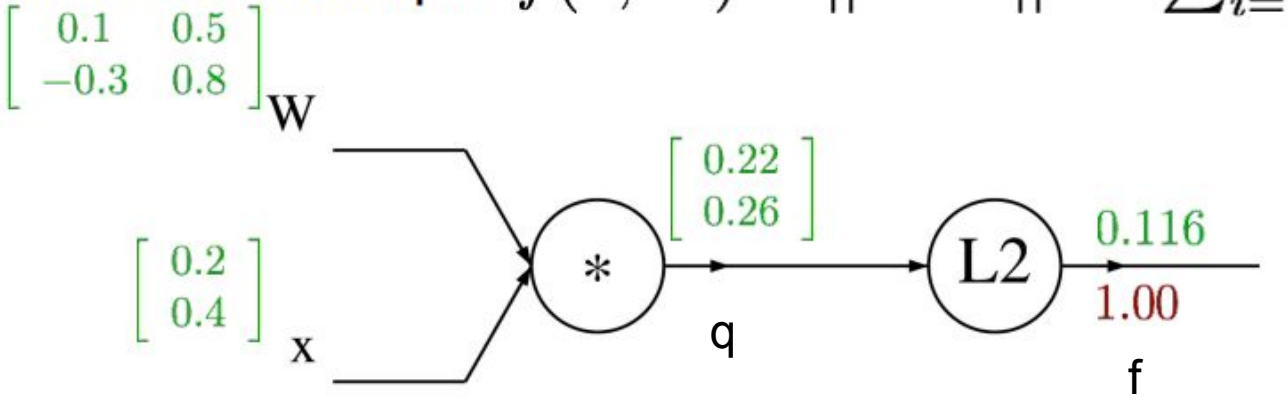
Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



Um exemplo vetorizado

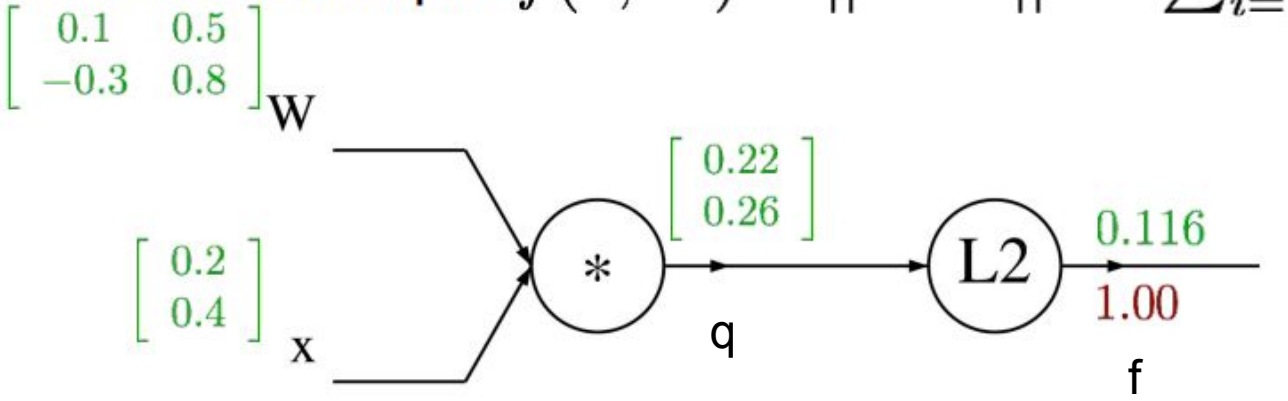
A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



$$f(q) = \left\| q^2 \right\| = q_1^2 + q_2^2$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



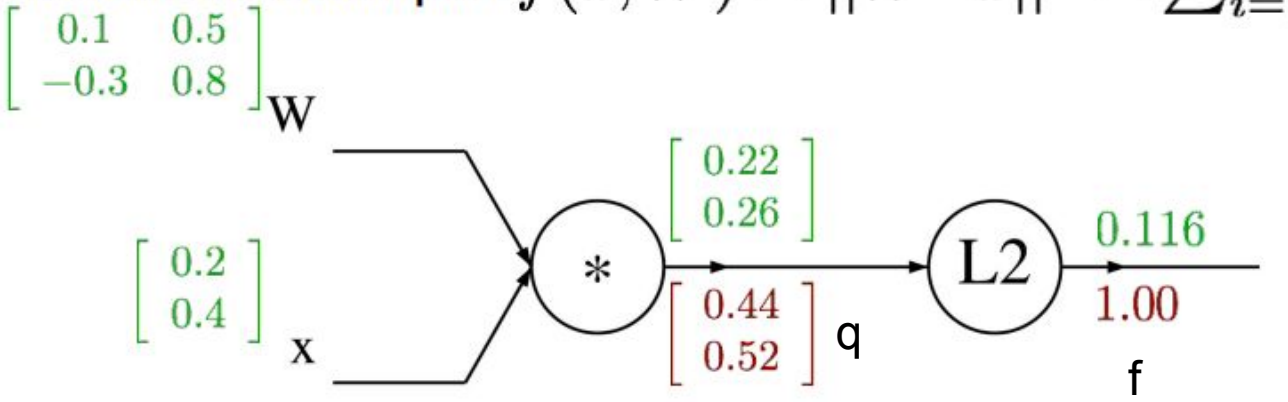
$$f(q) = \left\| q^2 \right\| = q_1^2 + q_2^2$$

$$\frac{\partial f}{\partial q_i} = 2q_i$$

$$\nabla_q f = 2q$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



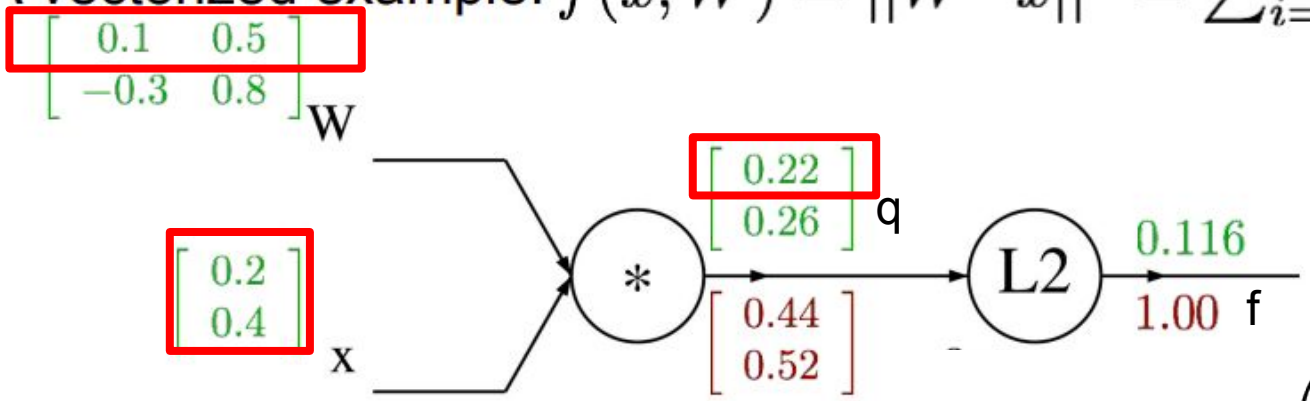
$$f(q) = \left\| q^2 \right\| = q_1^2 + q_2^2$$

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Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



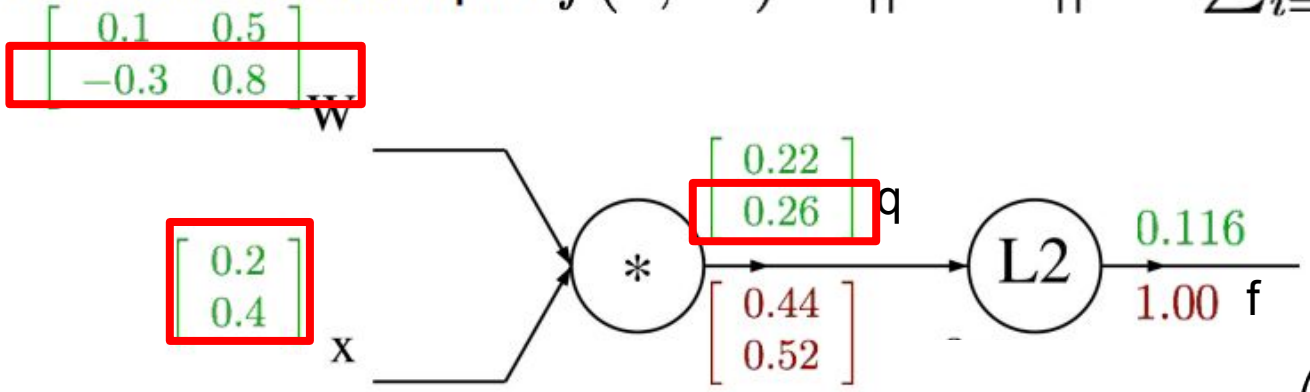
$$q = Wx = \begin{pmatrix} w_{1,1} * x_1 + w_{1,2} * x_2 \\ w_{2,1} * x_1 + w_{2,2} * x_2 \end{pmatrix}$$

$$\frac{\partial q_1}{\partial w_{1,1}} = x_1 \qquad \frac{\partial q_1}{\partial w_{1,2}} = x_2$$

$$\frac{\partial q_1}{\partial w_{2,1}} = 0 \qquad \frac{\partial q_1}{\partial w_{2,2}} = 0$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



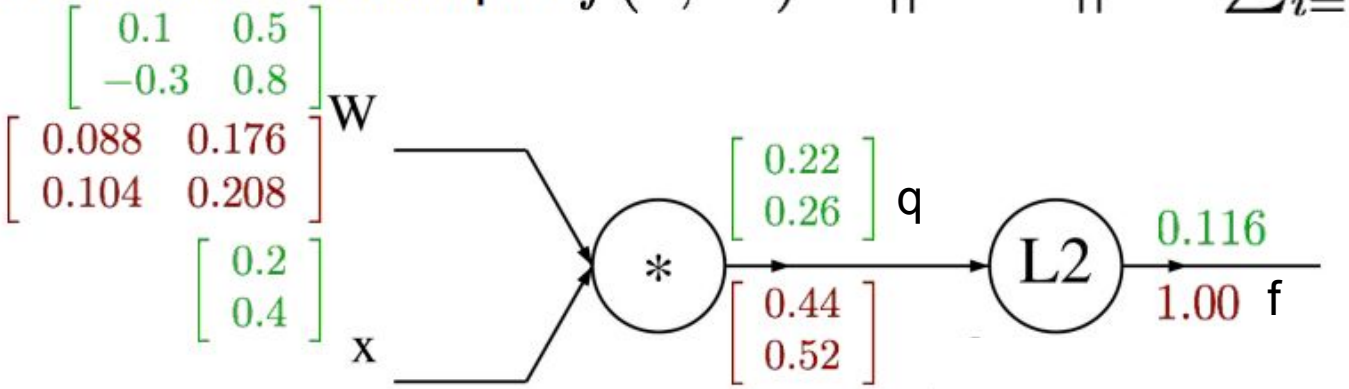
$$q = Wx = \begin{pmatrix} w_{1,1} * x_1 + w_{1,2} * x_2 \\ w_{2,1} * x_1 + w_{2,2} * x_2 \end{pmatrix}$$

$$\frac{\partial q_1}{\partial w_{1,1}} = x_1 \quad \frac{\partial q_1}{\partial w_{1,2}} = x_2 \quad \frac{\partial q_2}{\partial w_{1,1}} = 0 \quad \frac{\partial q_2}{\partial w_{1,2}} = 0$$

$$\frac{\partial q_1}{\partial w_{2,1}} = 0 \quad \frac{\partial q_1}{\partial w_{2,2}} = 0 \quad \frac{\partial q_2}{\partial w_{2,1}} = x_1 \quad \frac{\partial q_2}{\partial w_{2,2}} = x_2$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

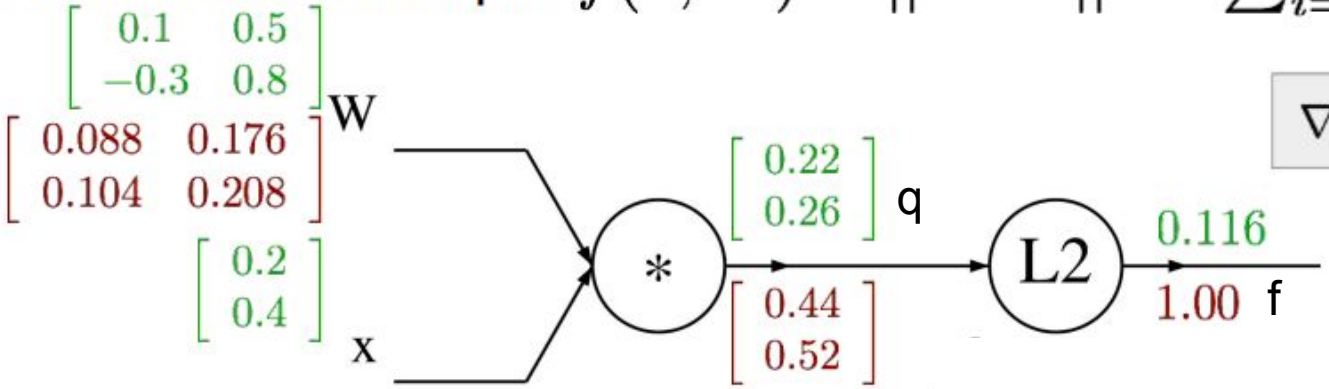


$$\frac{\partial q_k}{\partial W_{i,j}} = \mathbf{1}_{k=i} x_j$$

$$\begin{aligned}
\frac{\partial f}{\partial W_{i,j}} &= \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial W_{i,j}} \\
&= \sum (2q_k)(\mathbf{1}_{k=i} x_j) \\
&= 2q_i x_j
\end{aligned}$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



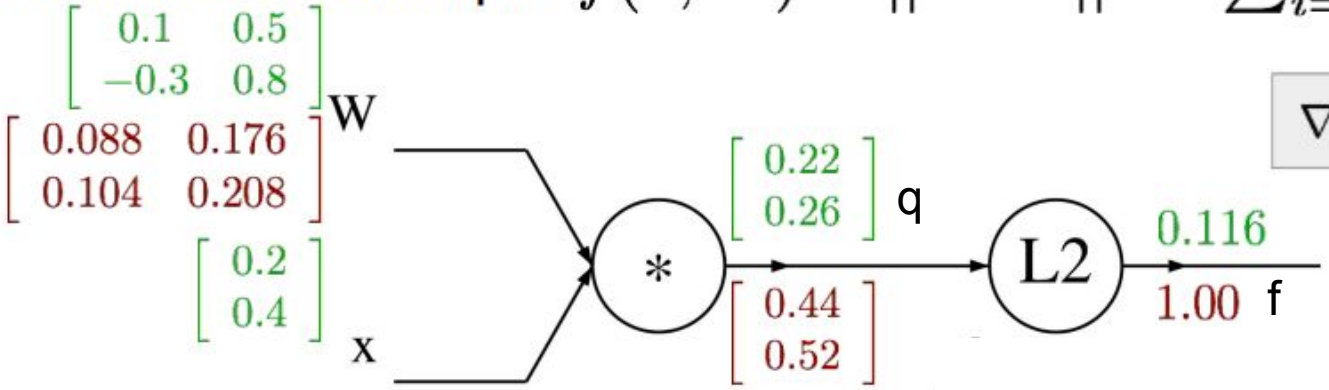
$$\nabla_W f = 2q \cdot x^T$$

$$\frac{\partial q_k}{\partial W_{i,j}} = \mathbf{1}_{k=i} x_j$$

$$\begin{aligned} \frac{\partial f}{\partial W_{i,j}} &= \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial W_{i,j}} \\ &= \sum (2q_k)(\mathbf{1}_{k=i} x_j) \\ &= 2q_i x_j \end{aligned}$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



$$\nabla_W f = 2q \cdot x^T$$

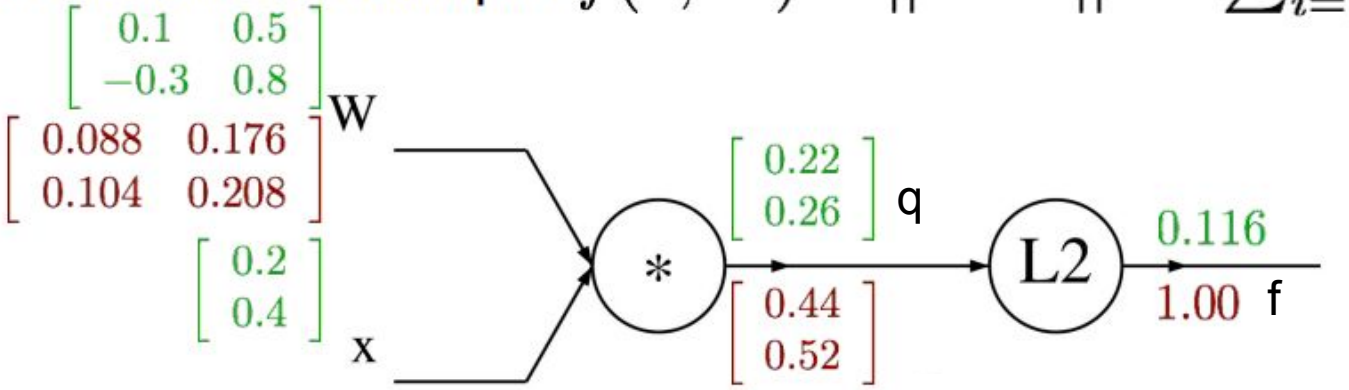
CONFIRA! O gradiente da variável sempre deve ter o mesmo "shape" da variável

$$\frac{\partial q_k}{\partial W_{i,j}} = \mathbf{1}_{k=i} x_j$$

$$\begin{aligned} \frac{\partial f}{\partial W_{i,j}} &= \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial W_{i,j}} \\ &= \sum (2q_k)(\mathbf{1}_{k=i} x_j) \\ &= 2q_i x_j \end{aligned}$$

Um exemplo vetorizado

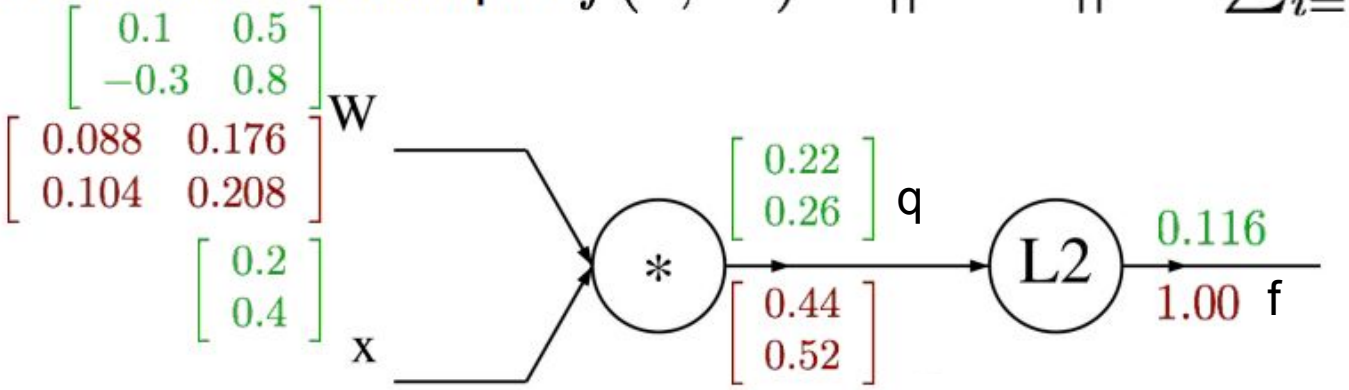
A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



$$q_1 = w_{1,1} * x_1 + w_{1,2} * x_2$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

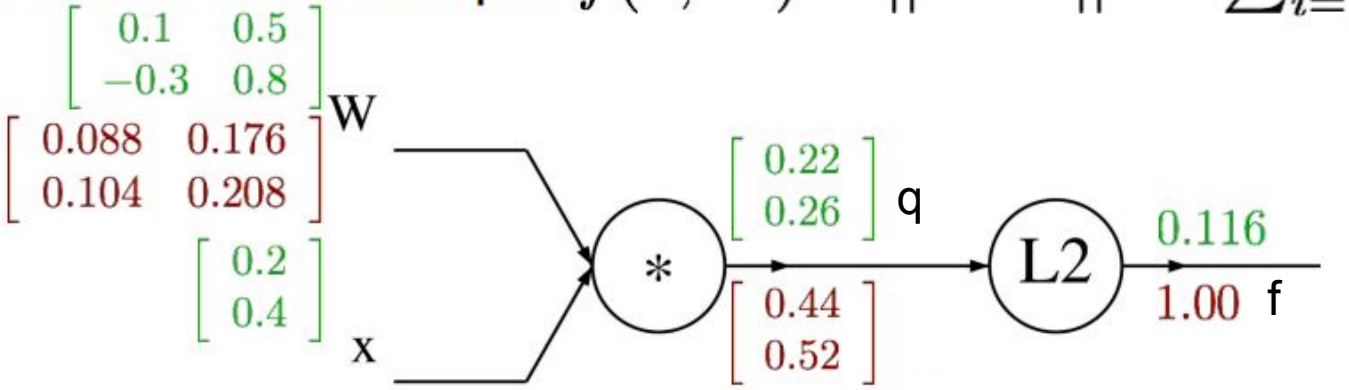


$$q_1 = w_{1,1} * x_1 + w_{1,2} * x_2$$

$$\frac{\partial q_1}{\partial x_1} = w_{1,1}$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

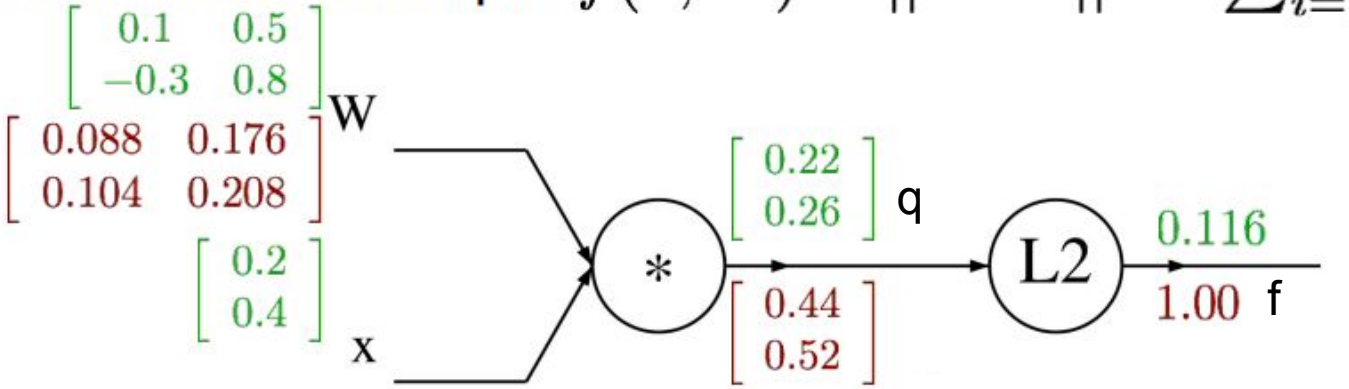


$$q_1 = w_{1,1} * x_1 + w_{1,2} * x_2$$

$$\frac{\partial q_1}{\partial x_1} = w_{1,1} \qquad \frac{\partial q_1}{\partial x_2} = w_{1,2}$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

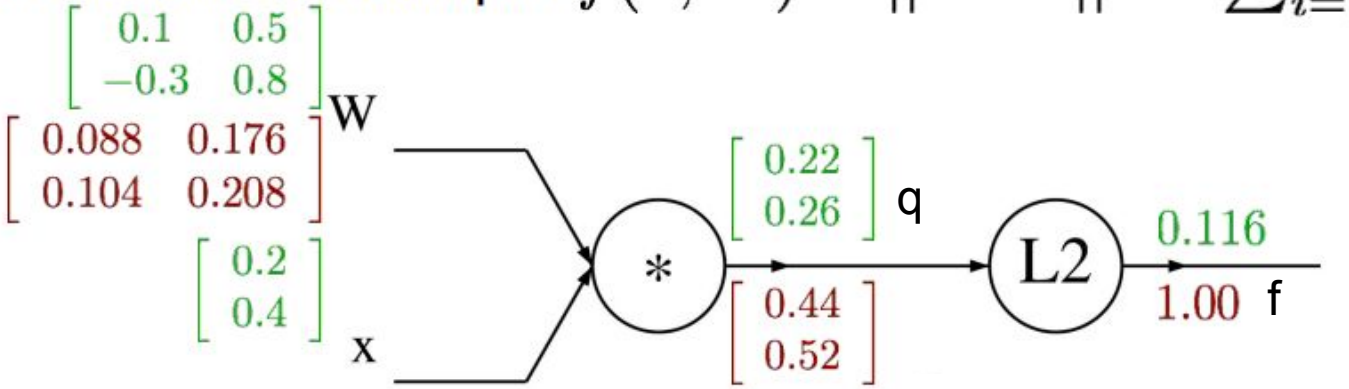


$$q_1 = w_{1,1} * x_1 + w_{1,2} * x_2 \qquad q_2 = w_{2,1} * x_1 + w_{2,2} * x_2$$

$$\frac{\partial q_1}{\partial x_1} = w_{1,1} \qquad \frac{\partial q_1}{\partial x_2} = w_{1,2}$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

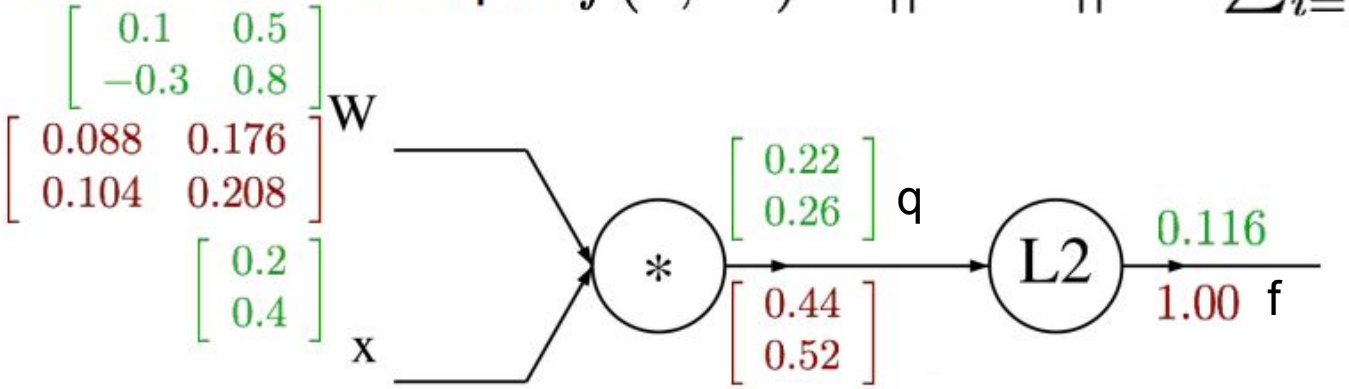


$$q_1 = w_{1,1} * x_1 + w_{1,2} * x_2 \qquad q_2 = w_{2,1} * x_1 + w_{2,2} * x_2$$

$$\frac{\partial q_1}{\partial x_1} = w_{1,1} \qquad \frac{\partial q_1}{\partial x_2} = w_{1,2} \qquad \frac{\partial q_2}{\partial x_1} = w_{2,1}$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

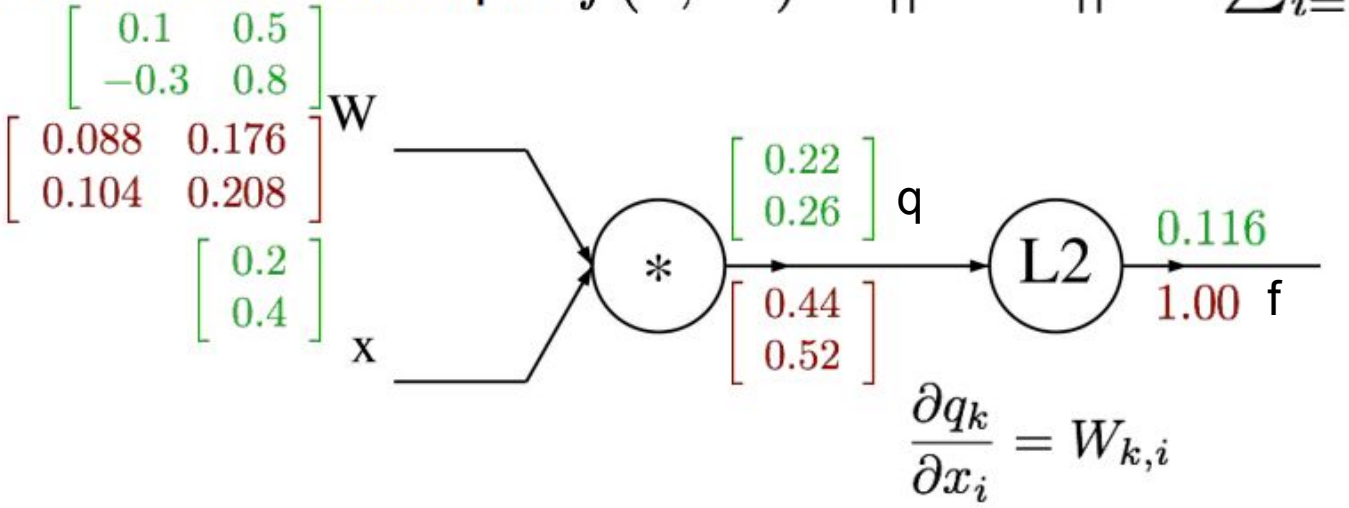


$$q_1 = w_{1,1} * x_1 + w_{1,2} * x_2 \qquad q_2 = w_{2,1} * x_1 + w_{2,2} * x_2$$

$$\frac{\partial q_1}{\partial x_1} = w_{1,1} \qquad \frac{\partial q_1}{\partial x_2} = w_{1,2} \qquad \frac{\partial q_2}{\partial x_1} = w_{2,1} \qquad \frac{\partial q_2}{\partial x_2} = w_{2,2}$$

Um exemplo vetorizado

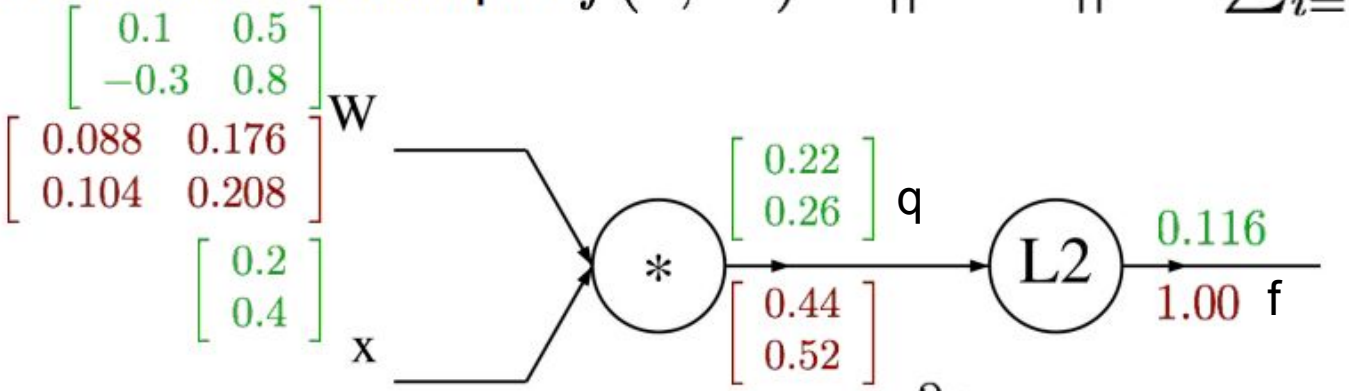
A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



$$\frac{\partial q_k}{\partial x_i} = W_{k,i}$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



$$\frac{\partial q_k}{\partial x_i} = W_{k,i}$$

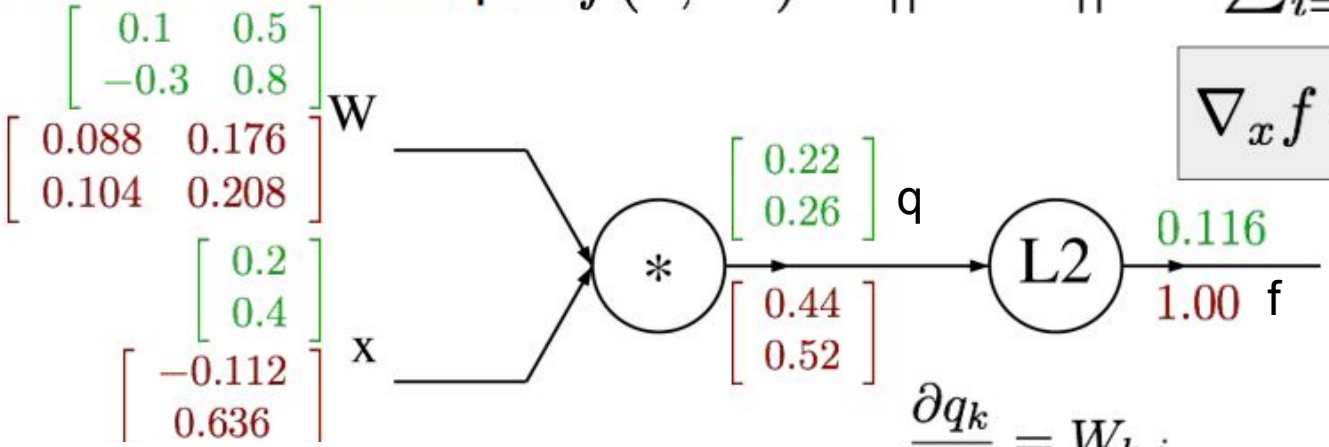
$$\frac{\partial f}{\partial x_i} = \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial x_i}$$

$$= \sum_k 2 q_k W_{k,i}$$

Um exemplo vetorizado

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

$$\nabla_x f = 2W^T \cdot q$$

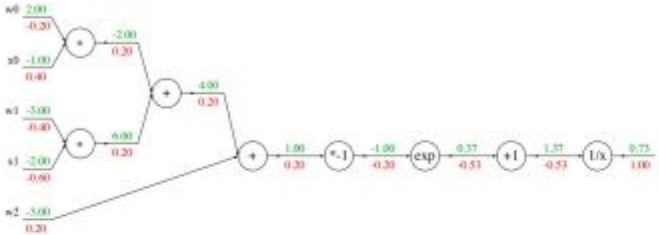


$$\frac{\partial q_k}{\partial x_i} = W_{k,i}$$

$$\frac{\partial f}{\partial x_i} = \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial x_i}$$

$$= \sum_k 2 q_k W_{k,i}$$

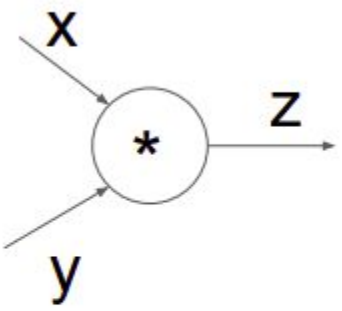
Implementação Modularizada: Forward / Backward API



Graph (or Net) object *(rough psuedo code)*

```
class ComputationalGraph(object):
    #...
    def forward(inputs):
        # 1. [pass inputs to input gates...]
        # 2. forward the computational graph:
        for gate in self.graph.nodes_topologically_sorted():
            gate.forward()
        return loss # the final gate in the graph outputs the loss
    def backward():
        for gate in reversed(self.graph.nodes_topologically_sorted()):
            gate.backward() # little piece of backprop (chain rule applied)
        return inputs_gradients
```

Implementação Modularizada: Forward / Backward API



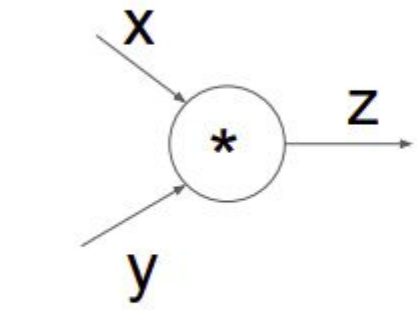
(x,y,z are scalars)

```
class MultiplyGate(object):
    def forward(x,y):
        z = x*y
        return z
    def backward(dz):
        # dx = ... #todo
        # dy = ... #todo
        return [dx, dy]
```

$\frac{\partial L}{\partial z}$

$\frac{\partial L}{\partial x}$

Implementação Modularizada: Forward / Backward API



(x,y,z are scalars)

```
class MultiplyGate(object):
    def forward(x,y):
        z = x*y
        self.x = x # must keep these around!
        self.y = y
        return z
    def backward(dz):
        dx = self.y * dz # [dz/dx * dL/dz]
        dy = self.x * dz # [dz/dy * dL/dz]
        return [dx, dy]
```

Atualizando os pesos

- No final, os pesos são atualizados

$$W = W - lr * \frac{\partial L}{\partial W}$$

Exercício

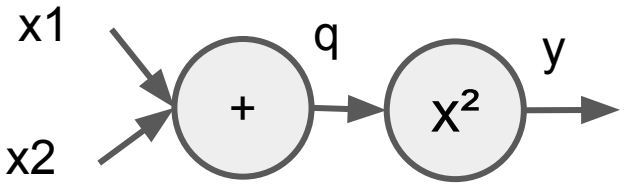
- Considere o seguinte grafo computacional descrito usando a biblioteca Pytorch, no Python (código no Classroom, aula_backpropagation_1.py).

```python
# Grafo Computacional
x1 = torch.tensor(3.0, requires_grad=True)  # Entrada 1
x2 = torch.tensor(4.0, requires_grad=True)  # Entrada 2

q = torch.add(x1, x2)
q.retain_grad()
y = torch.square(q)  # y = (x1 + x2)^2

print("y =", y.item())

y.backward()
print("dL/dq =", q.grad.item())
print("dL/dx1 =", x1.grad.item())
print("dL/dx2 =", x2.grad.item())
```



Exercício

- Considere o seguinte grafo computacional descrito usando a biblioteca Pytorch, no Python (código aula_backpropagation_1.py).

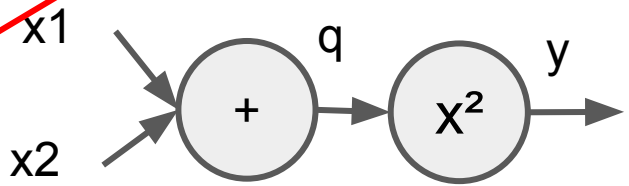
```
# Grafo Computacional
x1 = torch.tensor(3.0, requires_grad=True)  # Entrada 1
x2 = torch.tensor(4.0, requires_grad=True)  # Entrada 2

q = torch.add(x1, x2)
q.retain_grad()
y = torch.square(q)  # y = (x1 + x2)^2

print("y =", y.item())

y.backward()
print("dL/dq =", q.grad.item())
print("dL/dx1 =", x1.grad.item())
print("dL/dx2 =", x2.grad.item())
```

```
y = 49.0
dL/dq = 14.0
dL/dx1 = 14.0
dL/dx2 = 14.0
```



Exercício

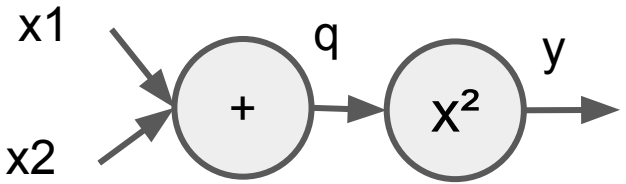
- Remova a operação nativa `torch.square(q)` e faça a sua implementação para elevar os 2 números ao quadrado, `Quadrado()`
-
-

```
# Grafo Computacional
x1 = torch.tensor(3.0, requires_grad=True)  # Entrada 1
x2 = torch.tensor(4.0, requires_grad=True)  # Entrada 2

q = torch.add(x1, x2)
q.retain_grad()
y = Quadrado.apply(q)

print("y =", y.item())

y.backward()
print("dL/dq =", q.grad.item())
print("dL/dx1 =", x1.grad.item())
print("dL/dx2 =", x2.grad.item())
```



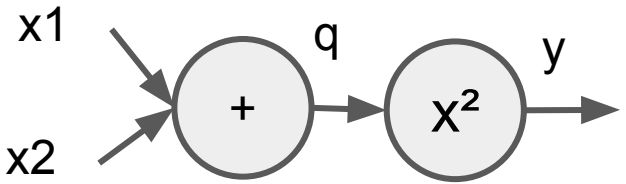
Exercício

- Remova a operação nativa `torch.square(q)` e faça a sua implementação para elevar os 2 números ao quadrado, `Quadrado()`
-
-

```python
class Quadrado(torch.autograd.Function):
    @staticmethod
    def forward(ctx, x):
        # salva valores necessários para o backward
        ctx.save_for_backward(x)

        # Todo: implementar a operação de quadrado
        # ...
        return y

    @staticmethod
    def backward(ctx, dL_dy):
        (x,) = ctx.saved_tensors
        # Todo: implementar a regra da cadeia para o quadrado
        #...
        return dL_dx
```

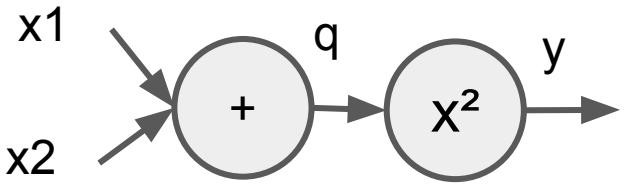


Exercício - Resolução

- Remova a operação nativa `torch.square(q)` e faça a sua implementação para elevar os 2 números ao quadrado, `Quadrado()`

```python
class Quadrado(torch.autograd.Function):
    @staticmethod
    def forward(ctx, x):
        # salva valores necessários para o backward
        ctx.save_for_backward(x)
        y = x ** 2
        return y

    @staticmethod
    def backward(ctx, dL_dy):
        (x,) = ctx.saved_tensors
        # dy/dx = 2x
        dL_dx = dL_dy * 2 * x
        return dL_dx
```



Backpropagation e como ele ajusta os pesos em redes profundas para minimizar o erro.

Davi Duarte de Paula

Fatec Rio Claro

Centro Paula Souza